

CSE 311:

Data Communication

Instructor:

Dr. Md. Monirul Islam

Analysis of Signals and Systems

Parseval's Theorem: Signal Energy in Frequency Domain

From Chapter 2,

Energy for real signal,

$$E_g = \int_{-\infty}^{\infty} g^2(t) dt$$

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Parseval's Theorem: Signal Energy in Frequency Domain

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$$\begin{aligned} g(t) &= \int_{-\infty}^{\infty} G(f)e^{j2\pi ft} df \\ g^*(t) &= \int_{-\infty}^{\infty} G(f)^* |e^{j2\pi ft}|^* df \\ &= \int_{-\infty}^{\infty} G(f)^* e^{-j2\pi ft} df \end{aligned}$$

Parseval's Theorem: Signal Energy in Frequency Domain

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$$= \int_{-\infty}^{\infty} g(t) \left[\int_{-\infty}^{\infty} G(f)^* e^{-j2\pi ft} df \right] dt$$

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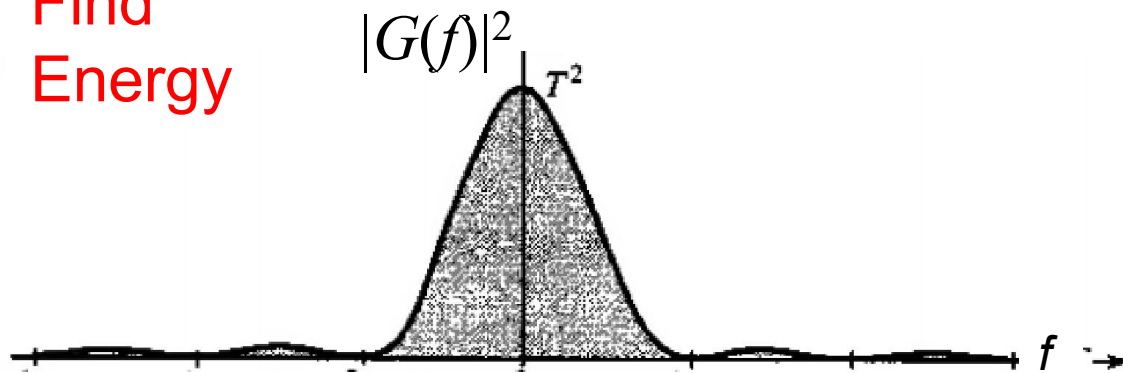
Parseval's Theorem: Signal Energy in Frequency Domain

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Application of Parseval's Theorem

$$E_g = \int_{-\infty}^{\infty} |g(t)|^2 dt = \int_{-\infty}^{\infty} |G(f)|^2 df$$

Find
Energy

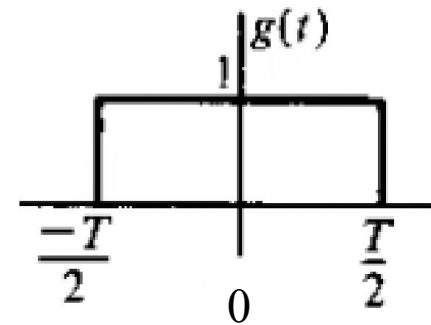
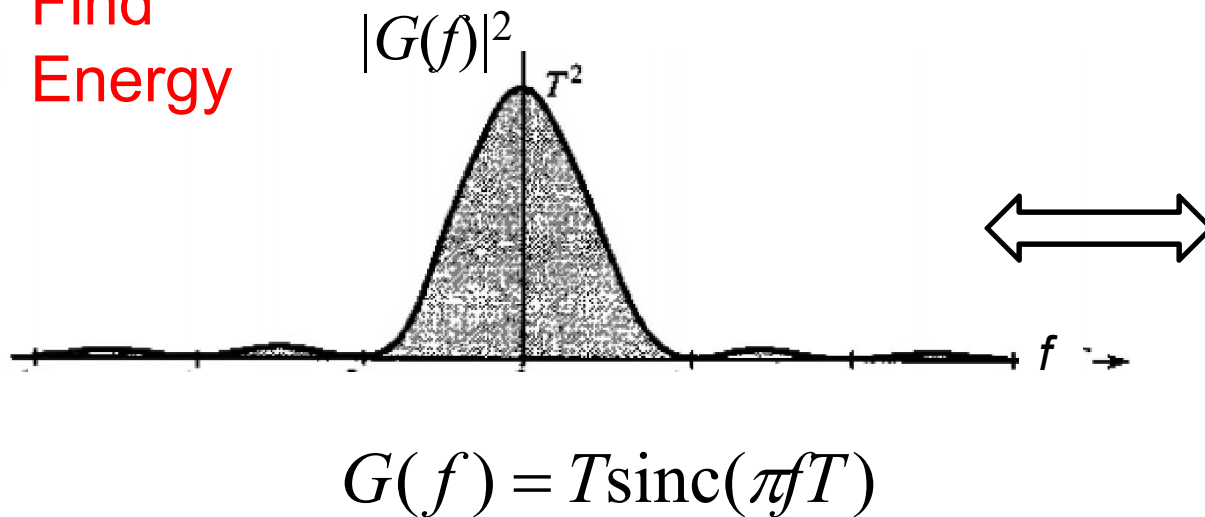


$$G(f) = T \text{sinc}(\pi f T)$$

Application of Parseval's Theorem

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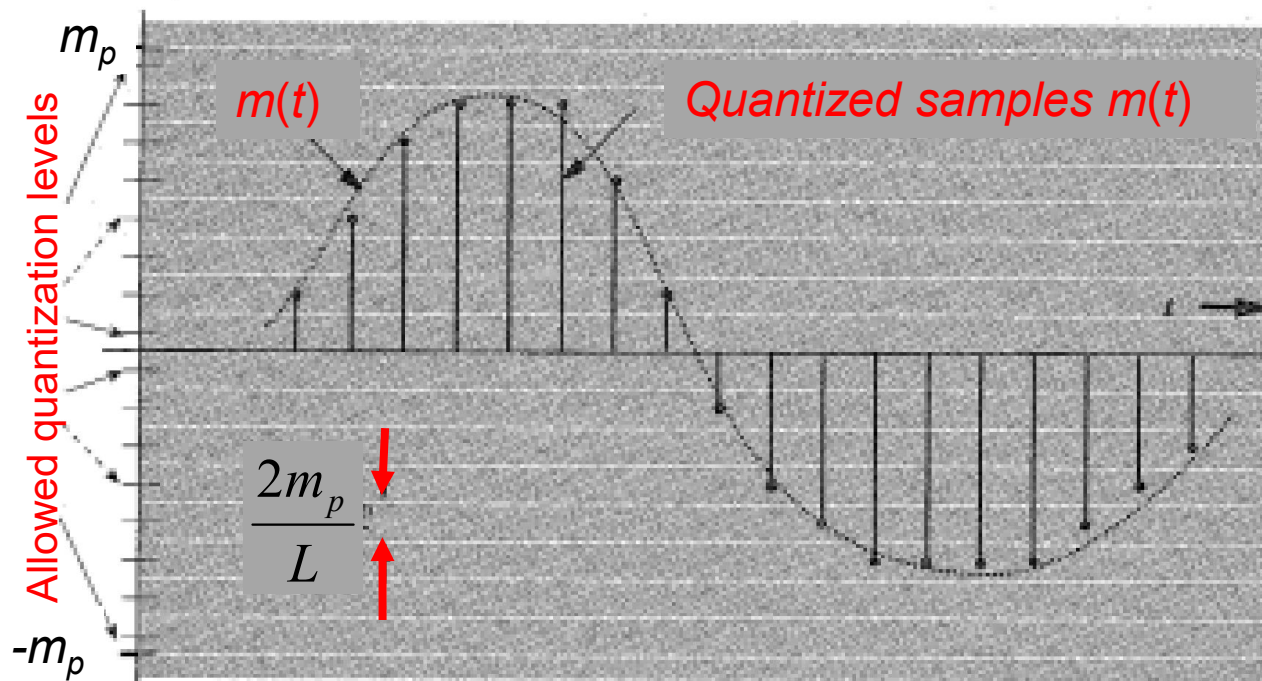
Find
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Sampling and Analog-to-Digital Conversion

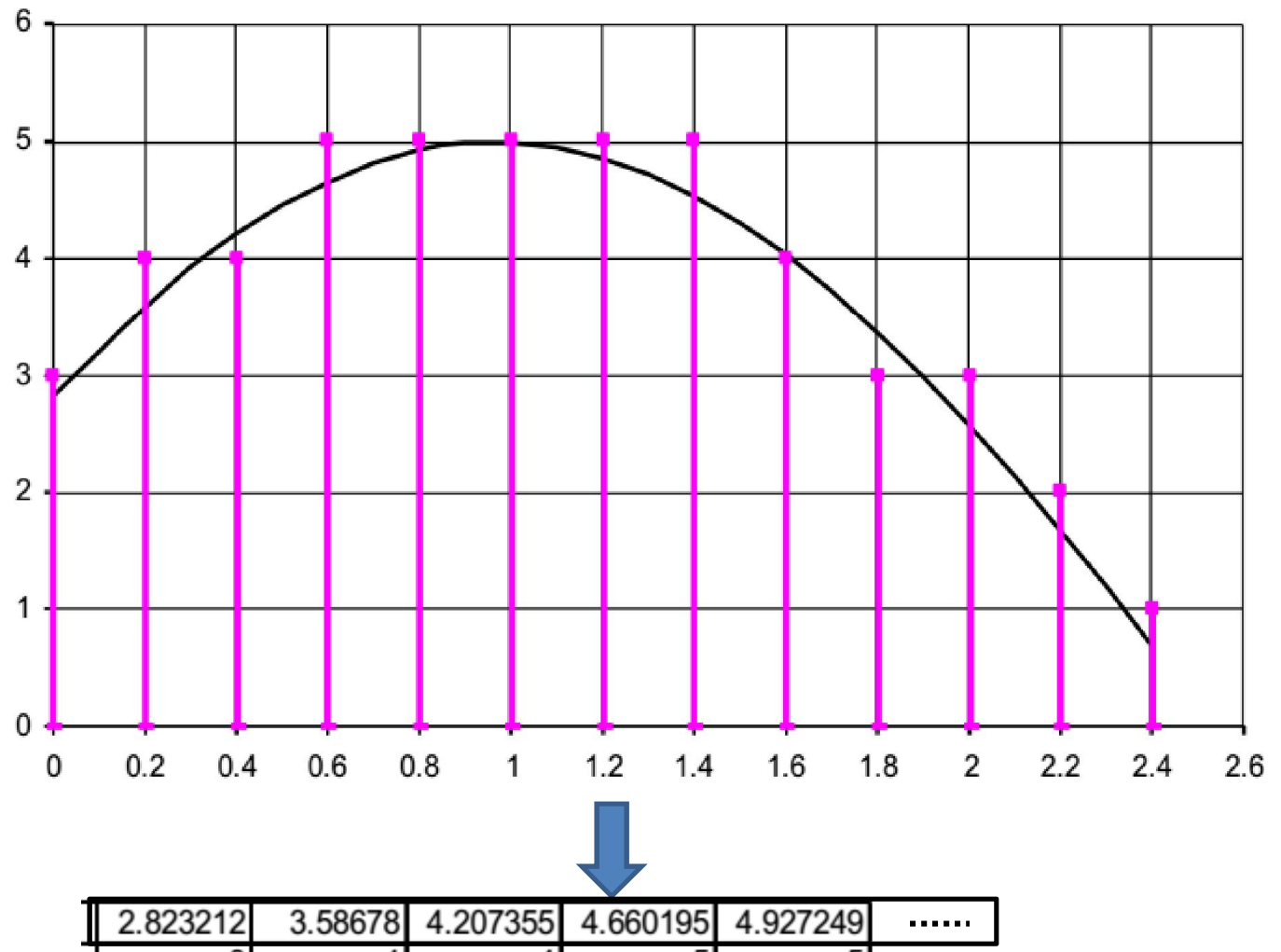
Analog to Digital Conversion of Message Signal

- 2 major steps
 - Sampling
 - Quantization



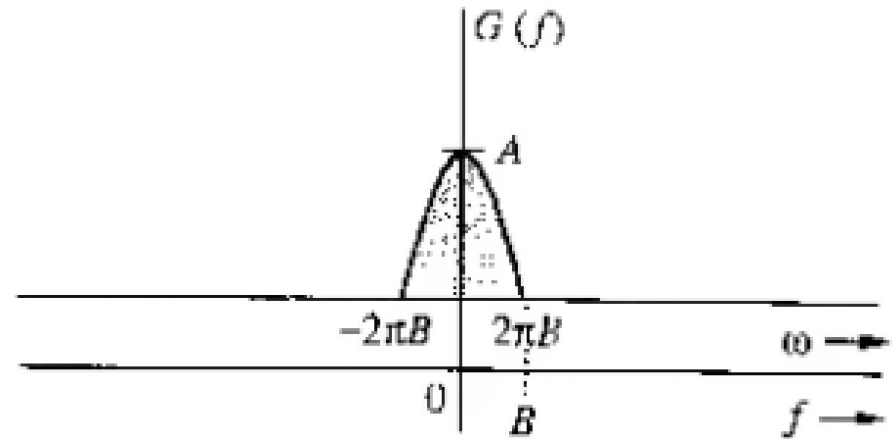
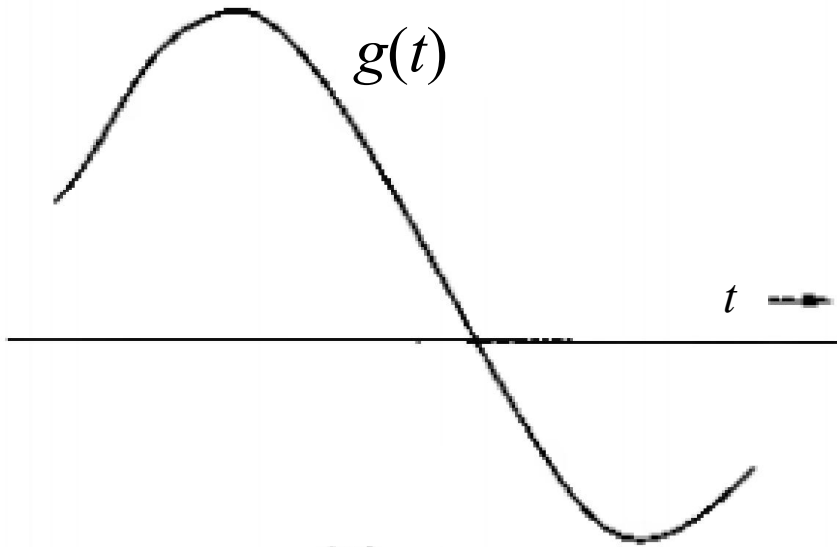
Analog to Digital Conversion of Message Signal

- Sampling



Sampling Theorem

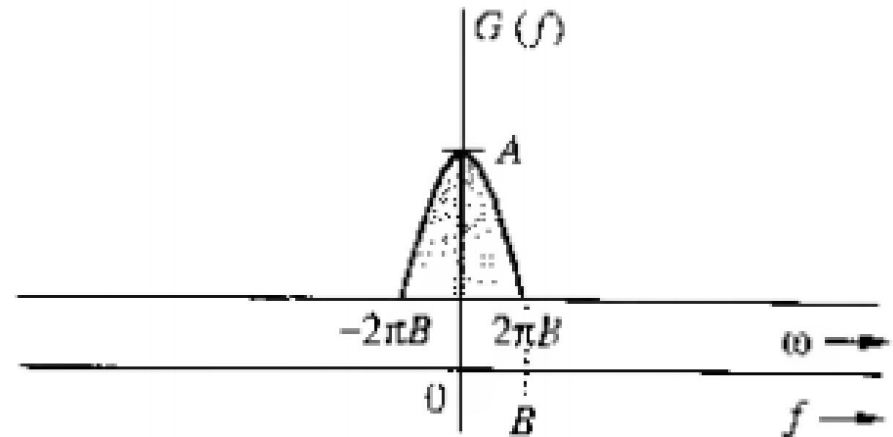
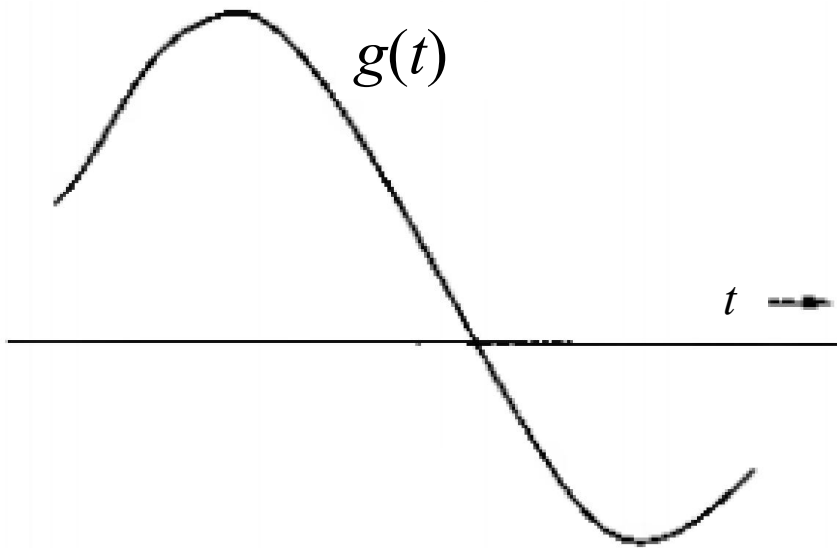
Let a message signal $g(t)$ band limited to B Hz



$$G(f) = 0 \text{ for all } |f| > B$$

Sampling Theorem

Let a message signal $g(t)$ band limited to B Hz

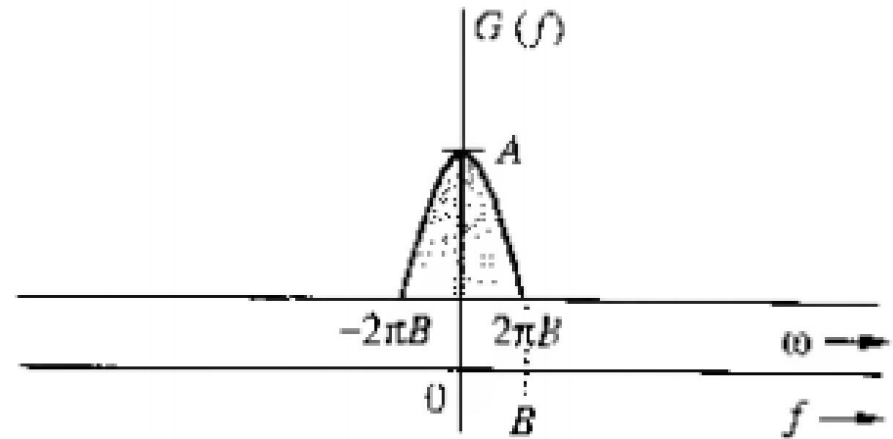
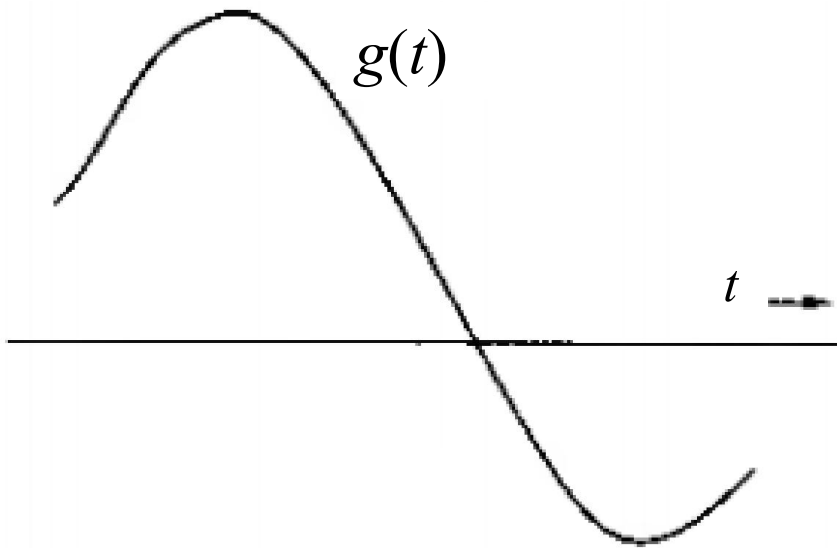


$$G(f) = 0 \text{ for all } |f| > B$$

The signal $g(t)$ can be reconstructed exactly from its discrete samples taken at a rate of f_s Hz, where $f_s \geq 2B$

Sampling Theorem

Let a message signal $g(t)$ band limited to B Hz

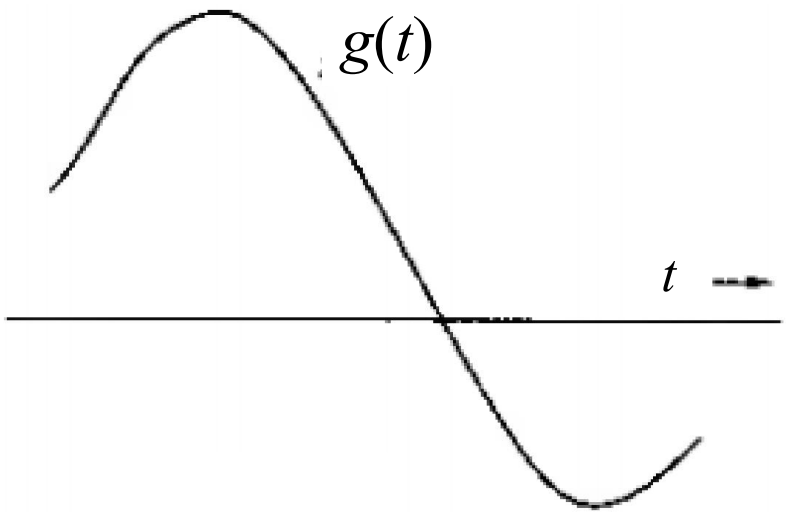


$$G(f) = 0 \text{ for all } |f| > B$$

For recovery, the minimum sampling $f_s = 2B$ Hz

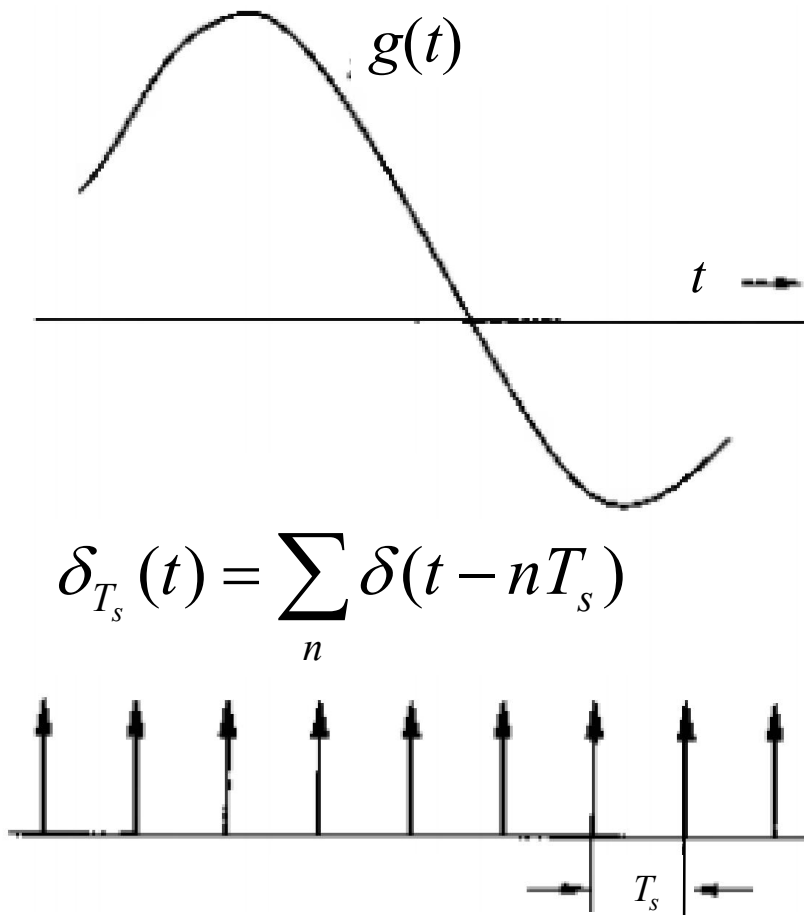
Sampling Theorem

Uniform sampling $g(t)$ at a rate f_s



Sampling Theorem

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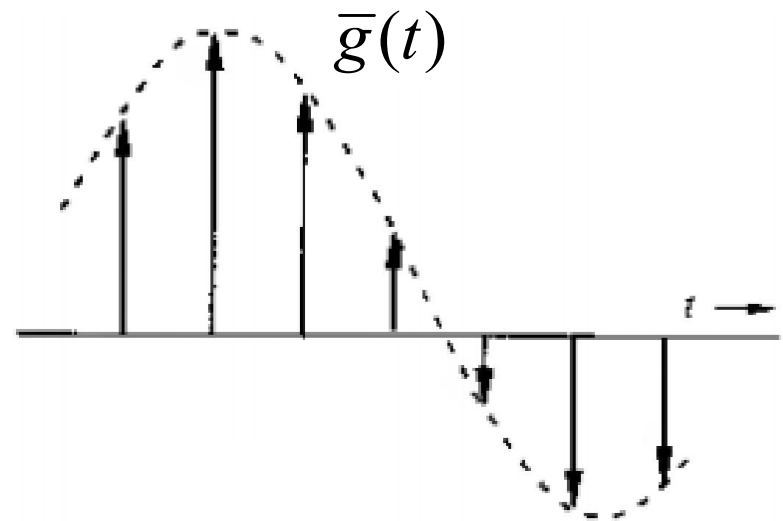
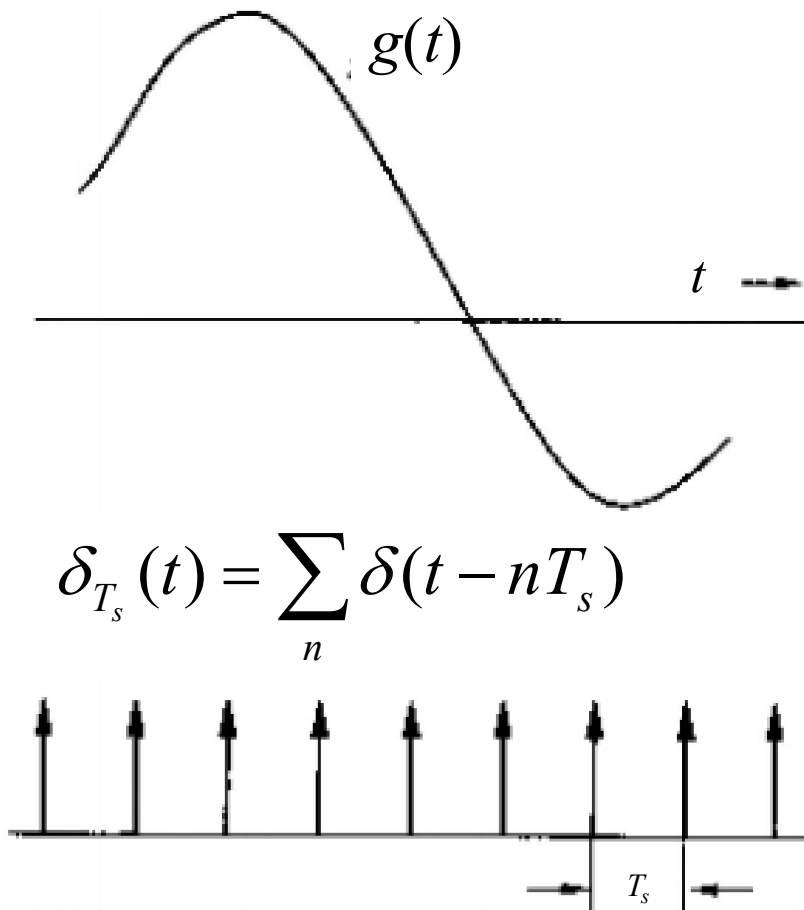
Multiply $g(t)$ by impulse train $\delta_{T_s}(t)$

where sampling period $T_s = 1/f_s$

The result is $\bar{g}(t)$

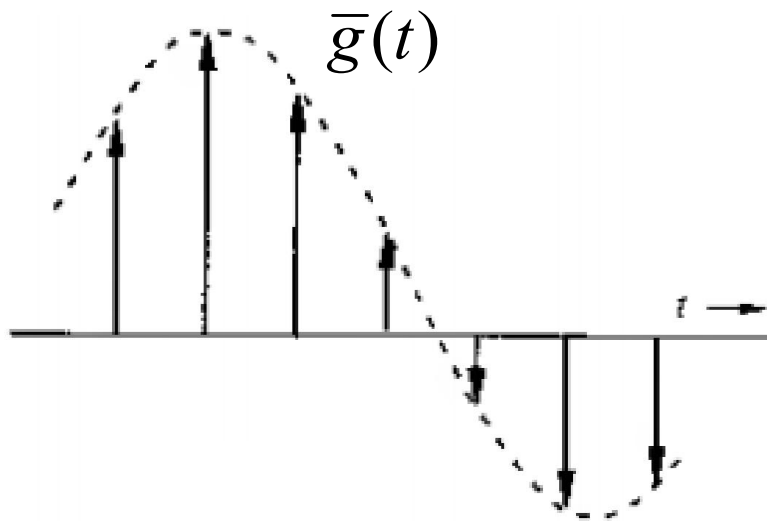
Sampling Theorem

Uniform sampling $g(t)$ at a rate f_s



Sampling Theorem

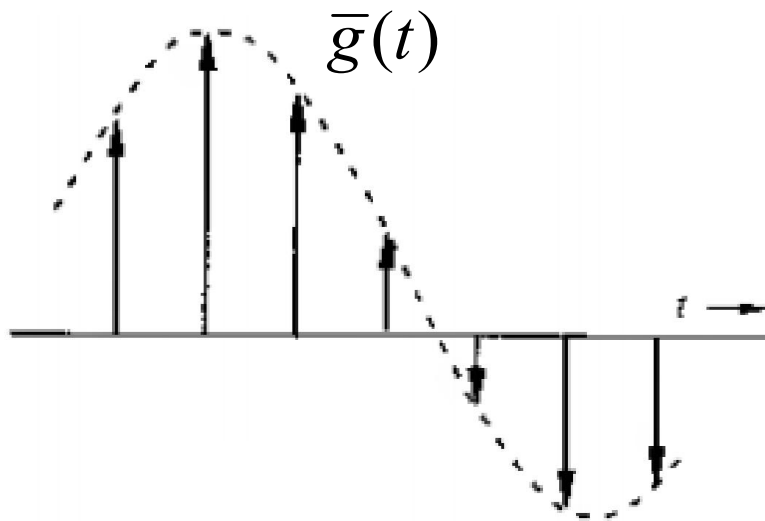
Uniform sampling $g(t)$ at a rate f_s



- $\bar{g}(t)$ consists of impulses uniformly spaced T_s seconds apart
- n th impulse is weighted by $g(nT_s)$, the value of $g(t)$ at $t = nT_s$

Sampling Theorem

Uniform sampling $g(t)$ at a rate f_s

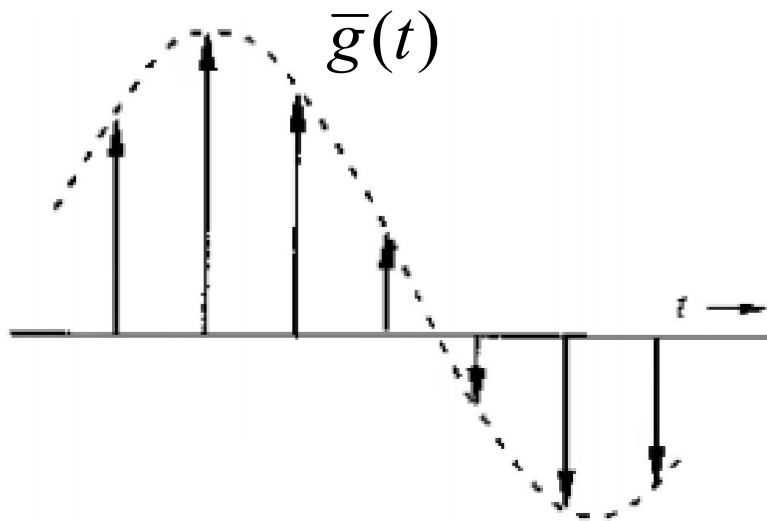


$\bar{g}(t)$ is multiplication of $g(t)$ and impulse train $\delta_{T_s}(t)$ and can be written as

$$\bar{g}(t) = g(t)\delta_{T_s}(t) = \sum_n g(nT_s)\delta(t - nT_s)$$

Sampling Theorem

Uniform sampling $g(t)$ at a rate f_s



$$\phi(t)\delta(t) = \phi(0)\delta(t)$$

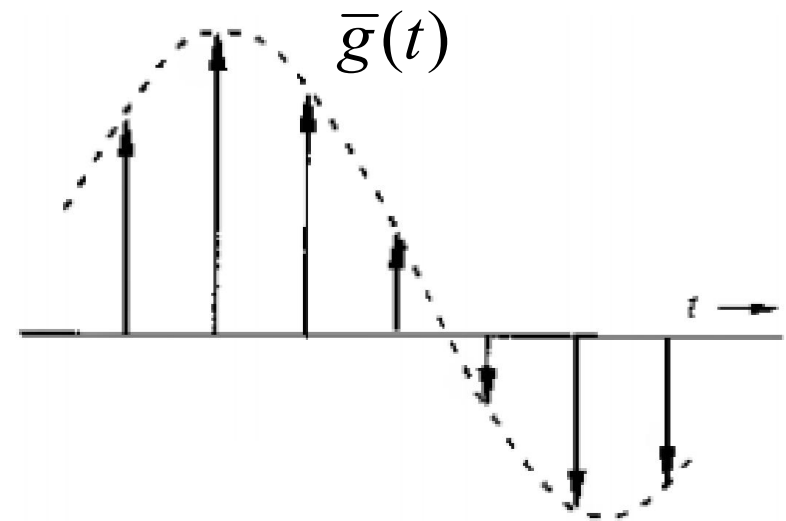
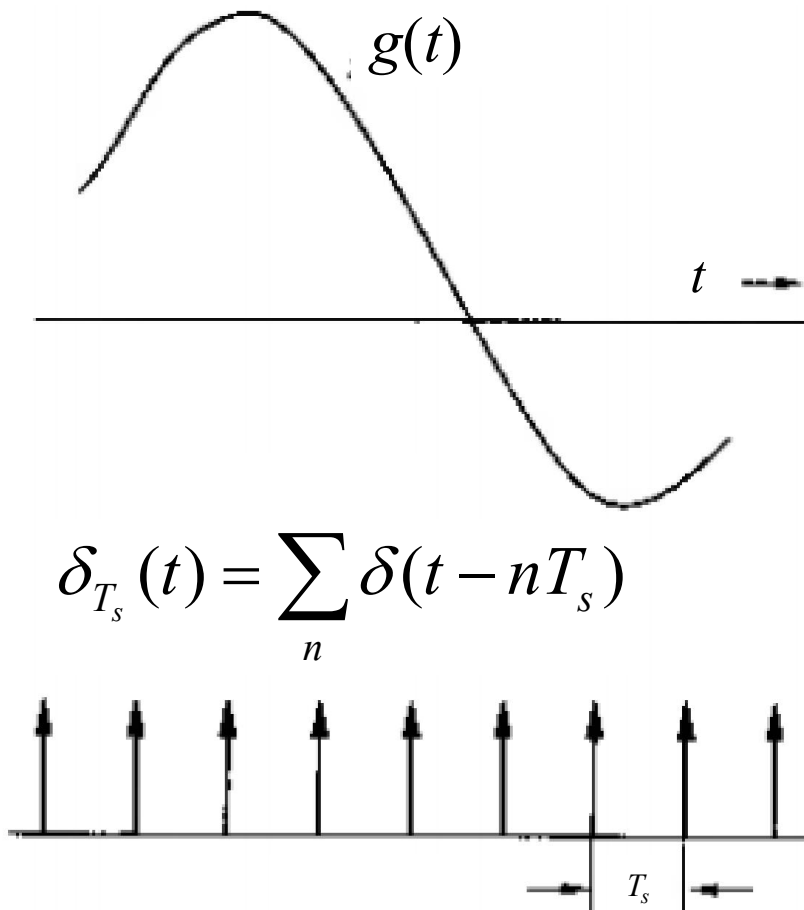
$$\phi(t)\delta(t - T) = \phi(T)\delta(t - T)$$

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Uniform sampling $g(t)$ at a rate $f_s = 1/T_s$

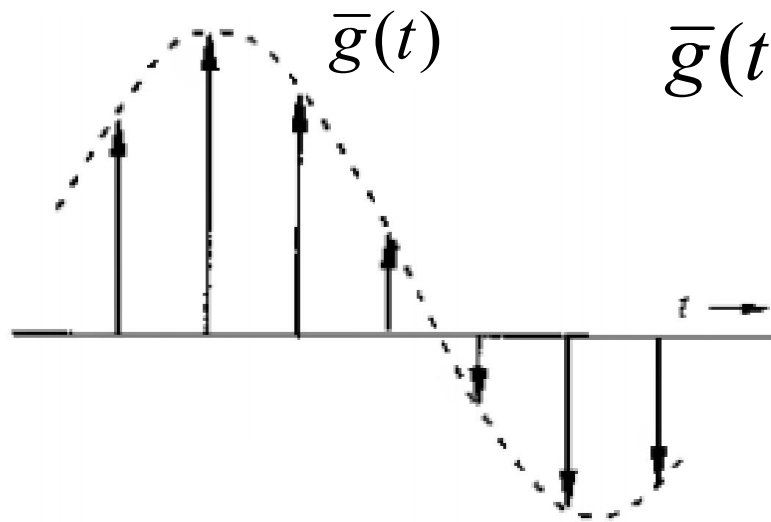


Alternate Representation

$$\delta_{T_s}(t) = \frac{1}{T_s} \sum_n e^{jn2\pi f_s t}$$

Sampling Theorem

Uniform sampling $g(t)$ at a rate $f_s = 1/T_s$



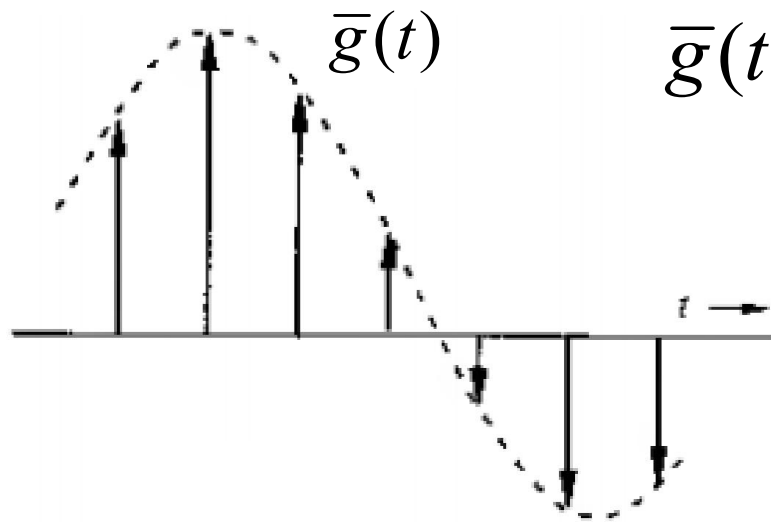
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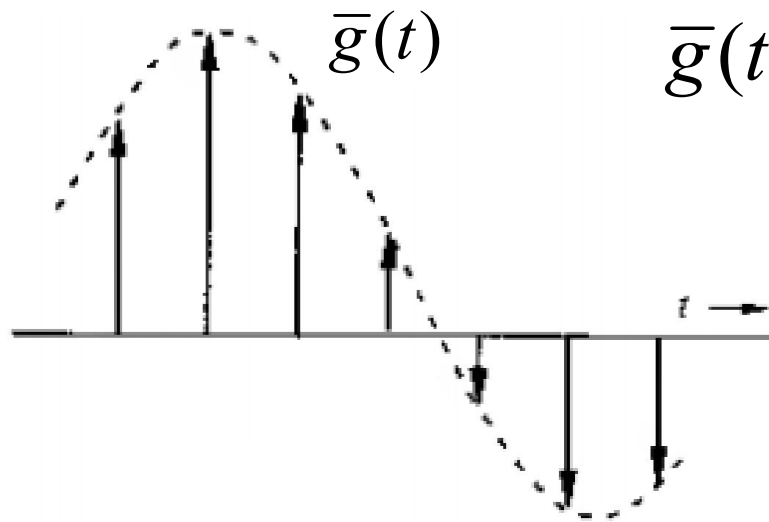
$$= \frac{1}{T_s} \sum_n g(t) e^{j2\pi n f_s t}$$

FT of $\bar{g}(t)$ is defined by

$$\bar{G}(f) = \mathfrak{F}[\bar{g}(t)] = \frac{1}{T_s} \sum_n G(f - nf_s)$$

Sampling Theorem

Uniform sampling $g(t)$ at a rate $f_s = 1/T_s$



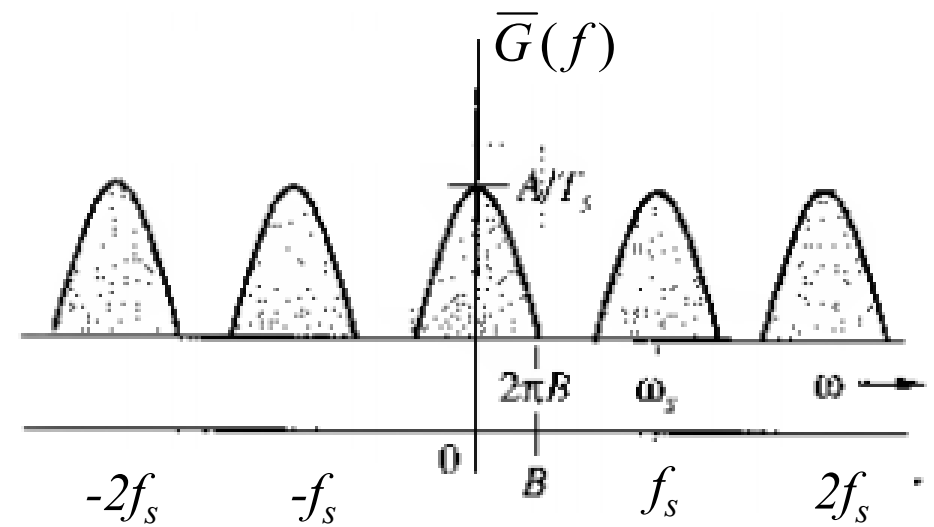
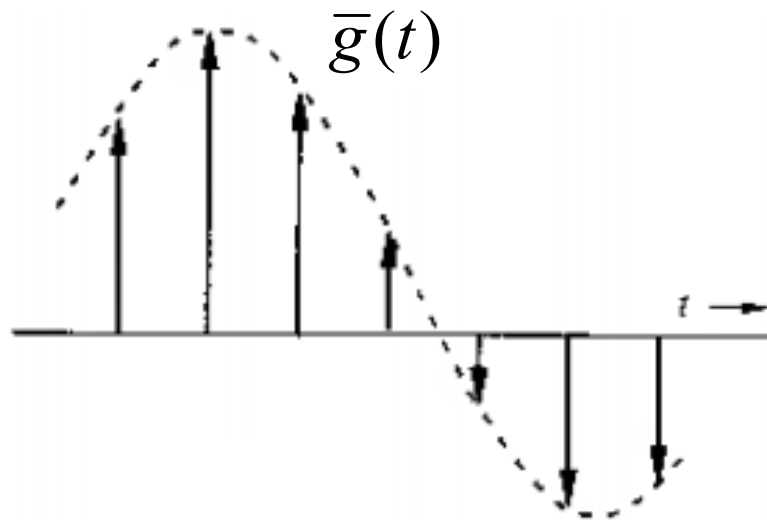
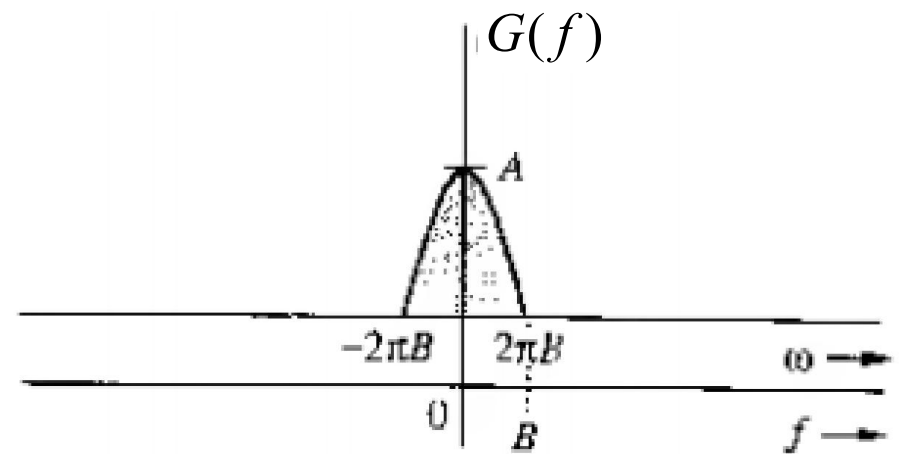
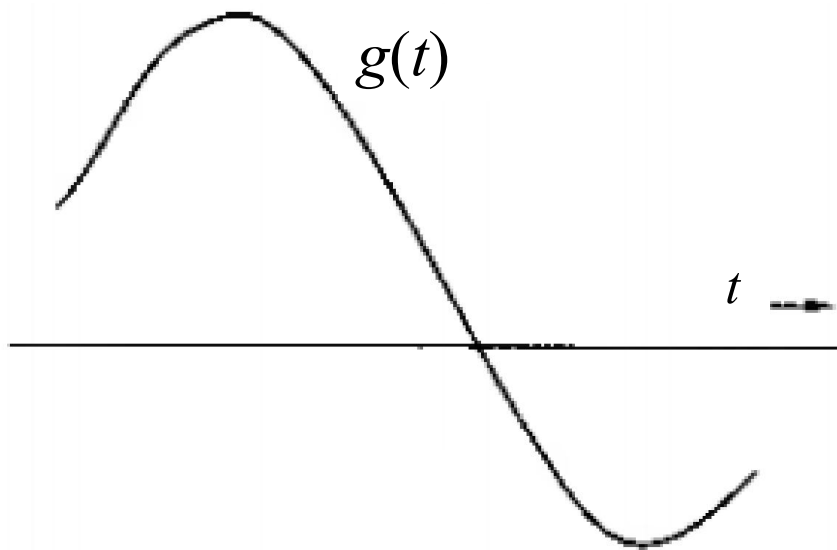
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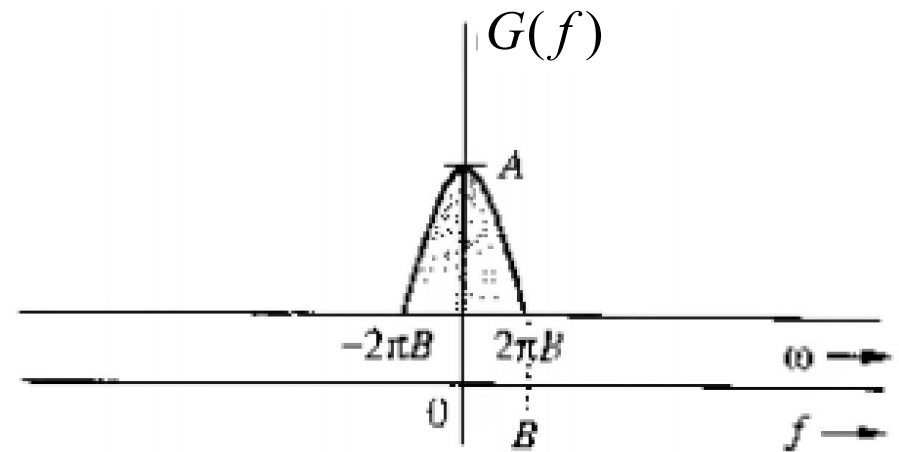
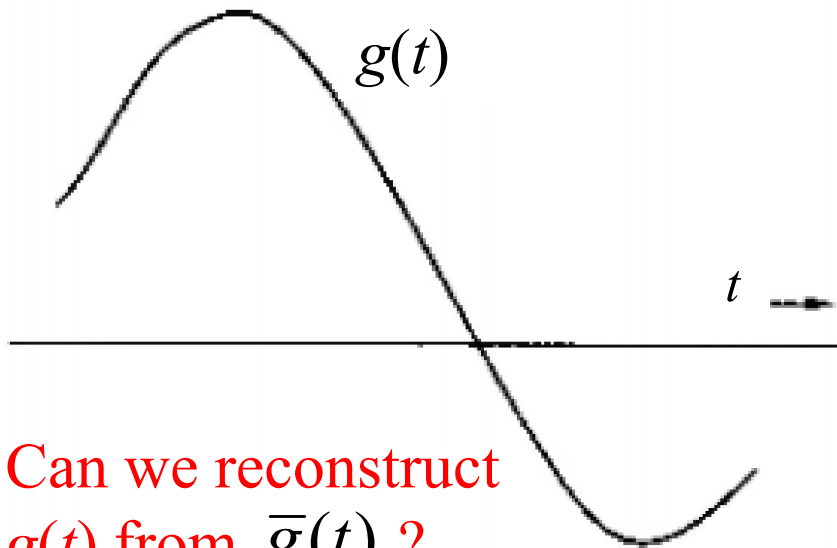
$$\bar{G}(f) = \mathfrak{F}[\bar{g}(t)] = \frac{1}{T_s} \sum_n G(f - nf_s)$$

$\bar{G}(f)$ consists of $G(f)$
scaled by f_s repeated
periodically with
period f_s

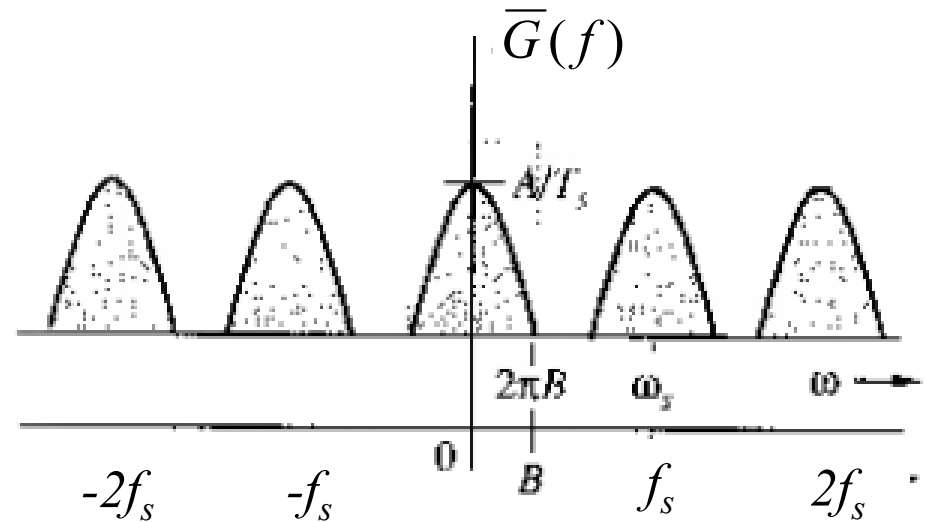
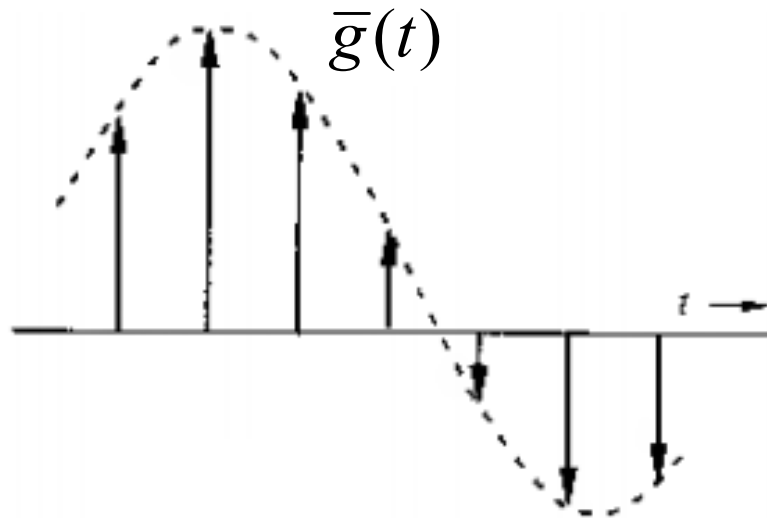
Sampling Theorem



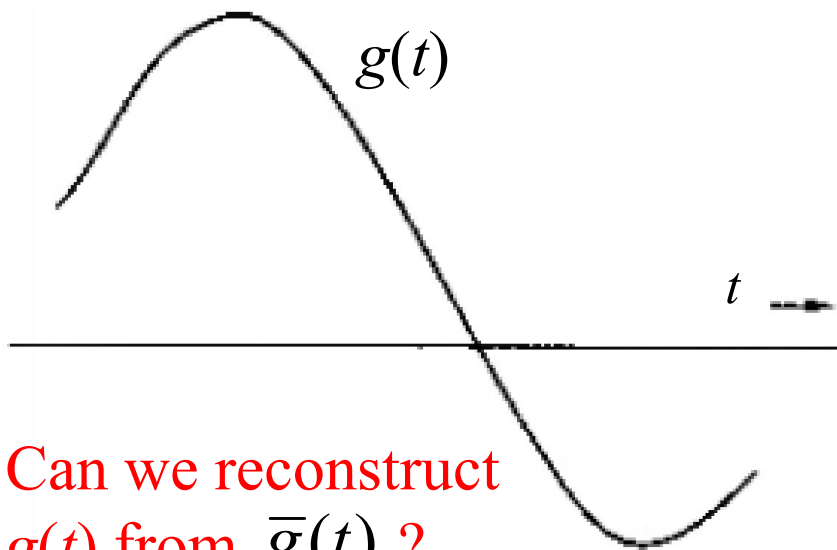
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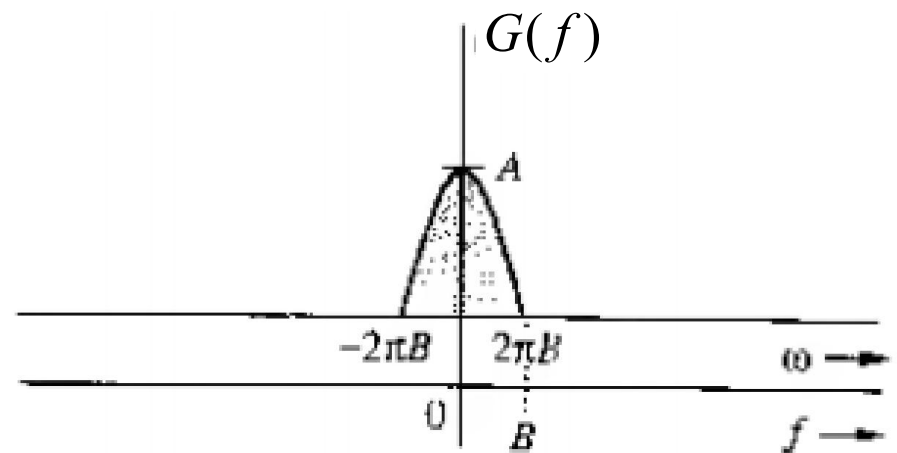
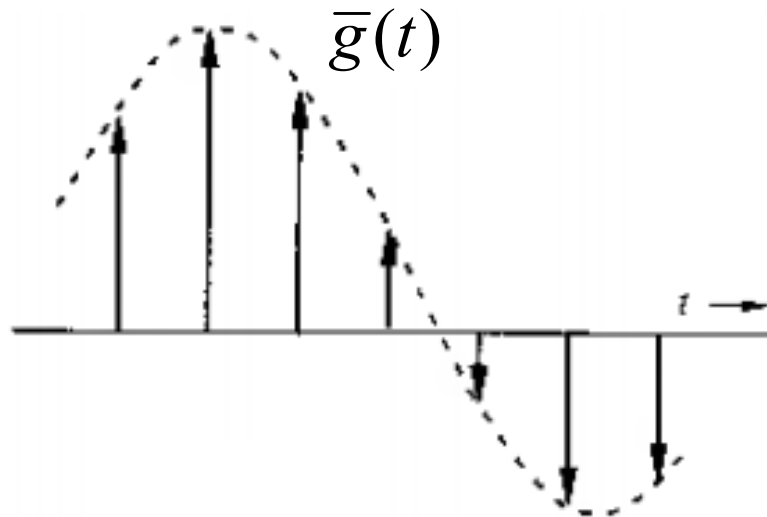
Can we reconstruct $g(t)$ from $\bar{g}(t)$?



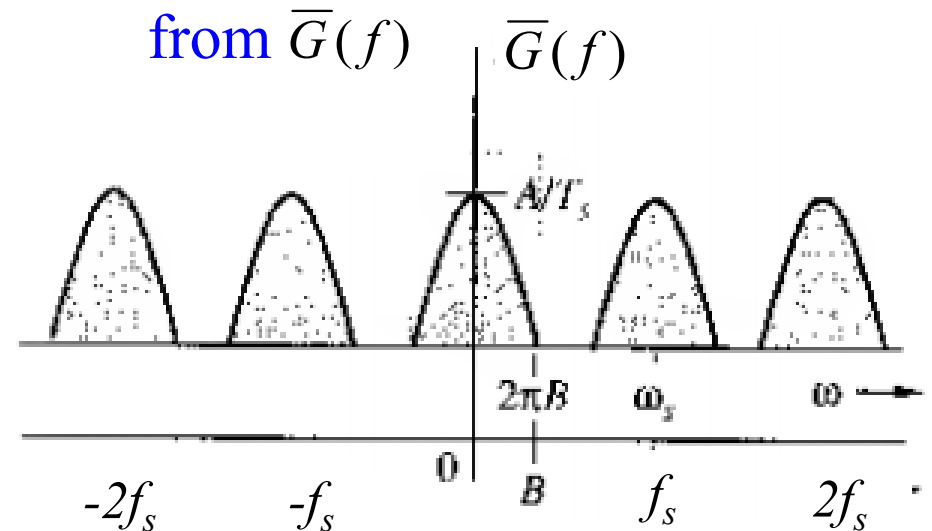
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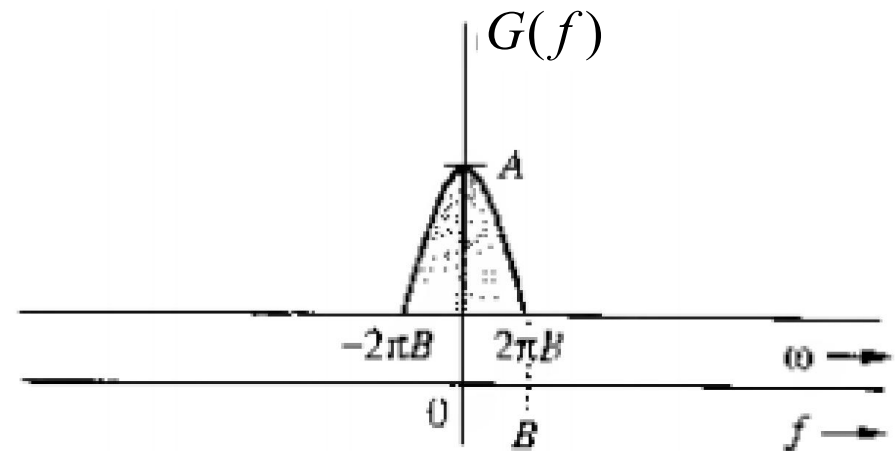
Possible, if we can
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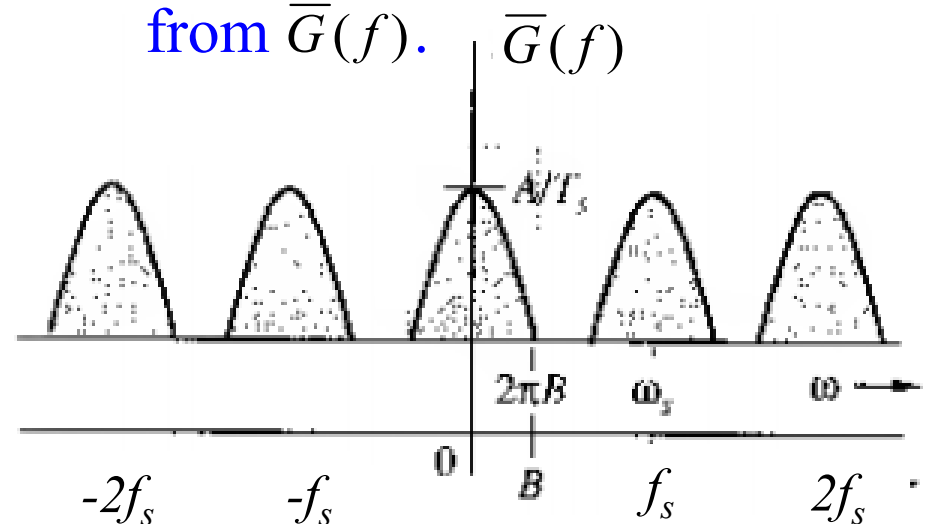
Sampling Theorem

- We have to isolate one cycle from $\bar{G}(f)$
- This is possible when cycles are non-overlapping, this means

$$f_s > 2B$$



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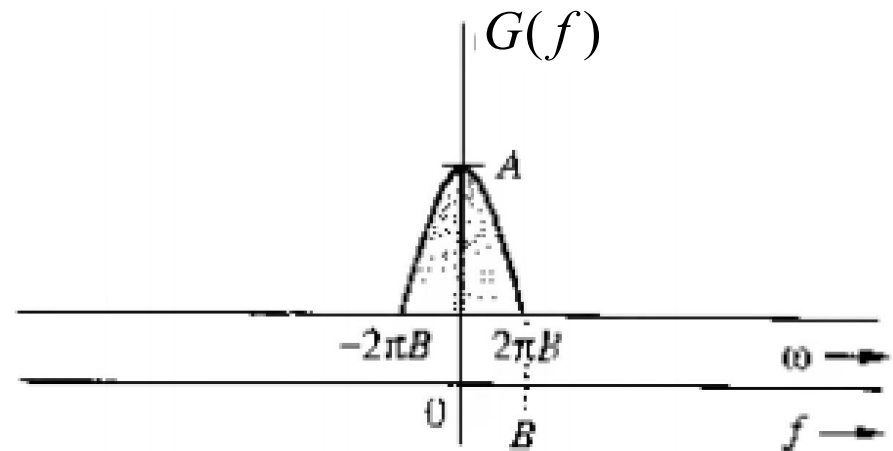
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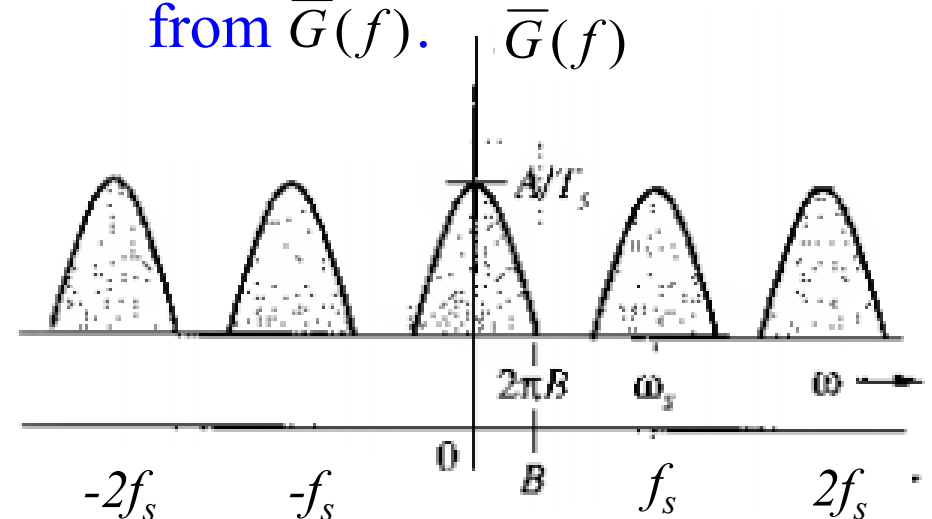
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Alternately,

Sampling interval of $g(t)$, $T_s < \frac{1}{2B}$



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Sampling Theorem

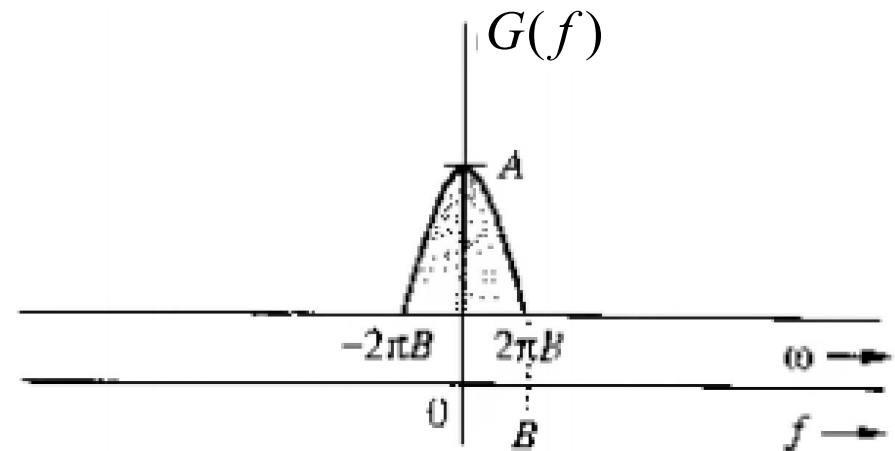
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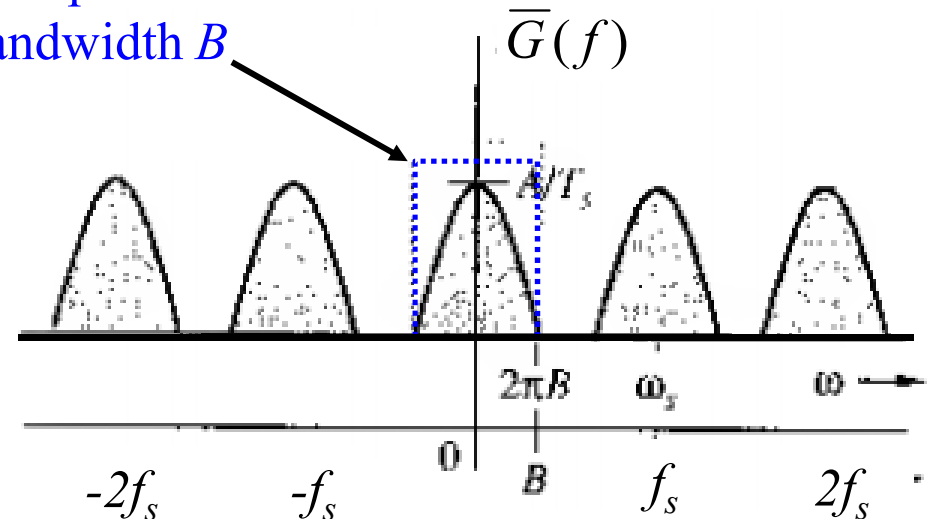
Alternately,

Sampling interval of $g(t)$, $T_s < \frac{1}{2B}$

Use an Ideal lowpass filter of bandwidth B to isolate one cycle.

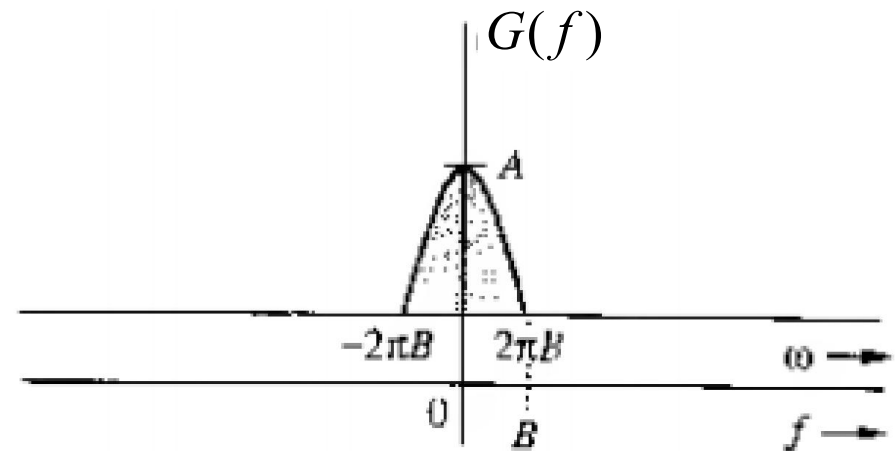


Low pass filter of bandwidth B



Sampling Theorem

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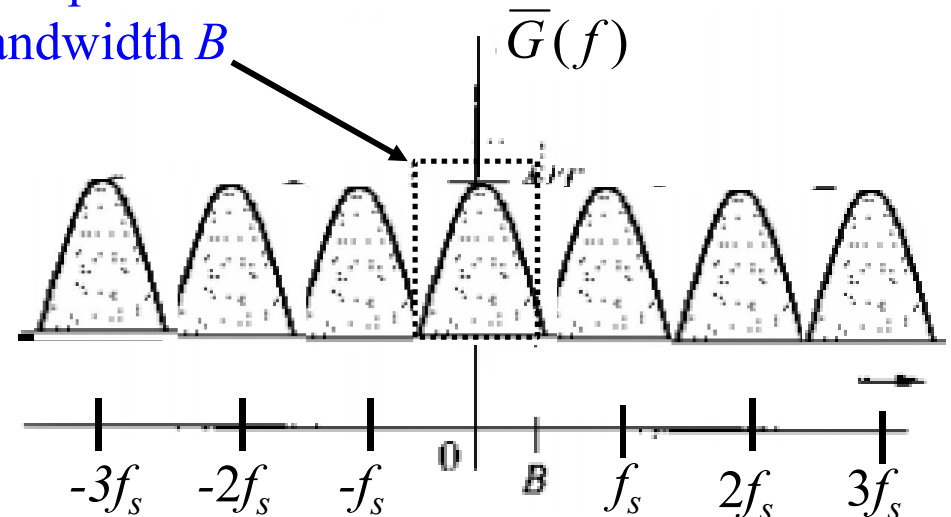


Minimum sampling rate can be

$$f_s = 2B$$

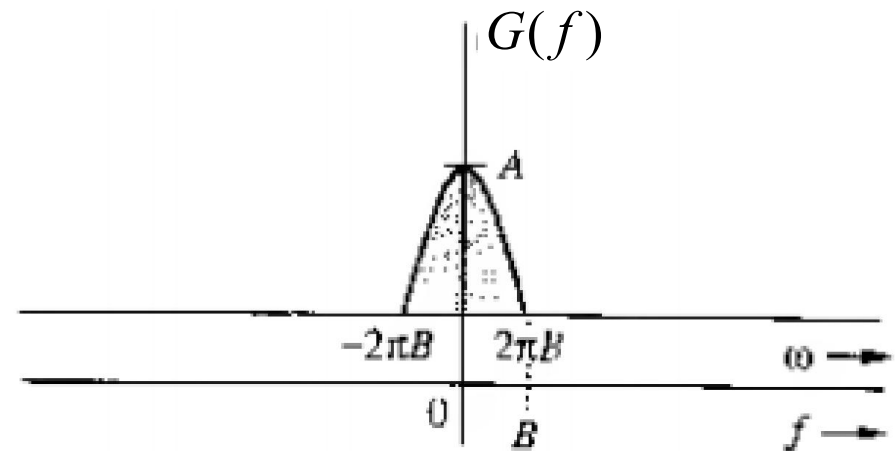
This is Nyquist sampling rate

Low pass filter of
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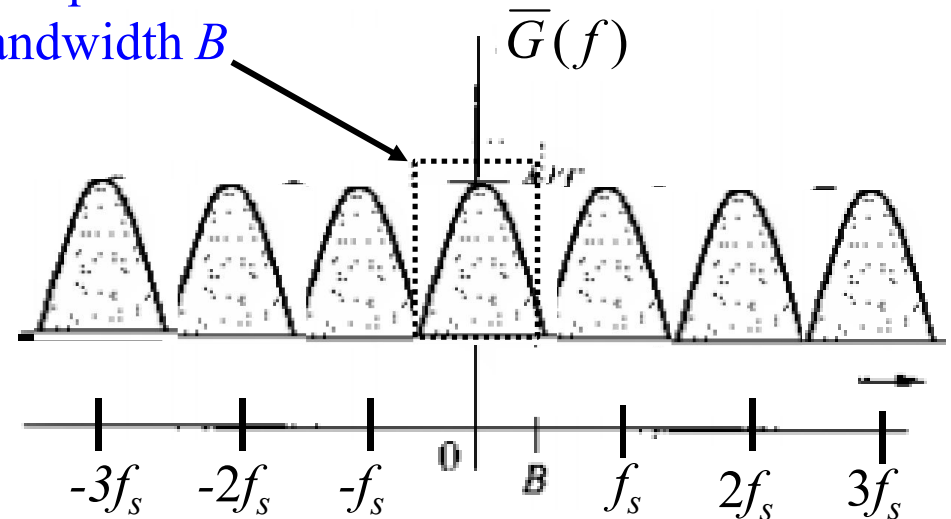


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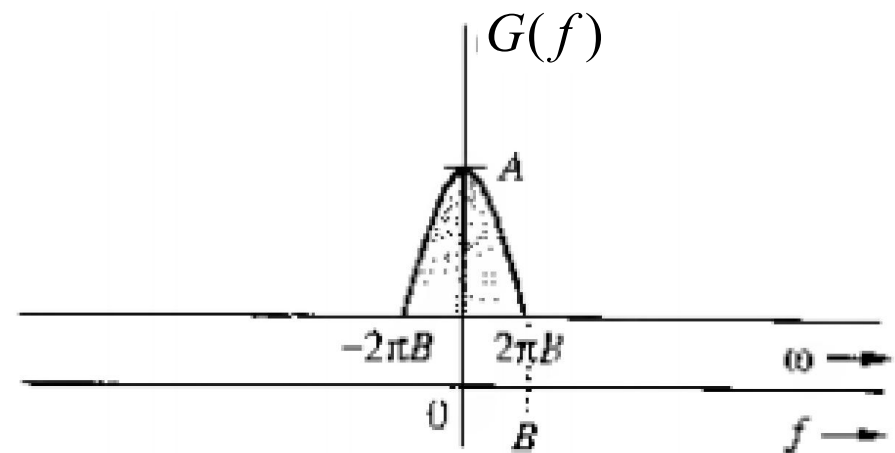
Low pass filter of
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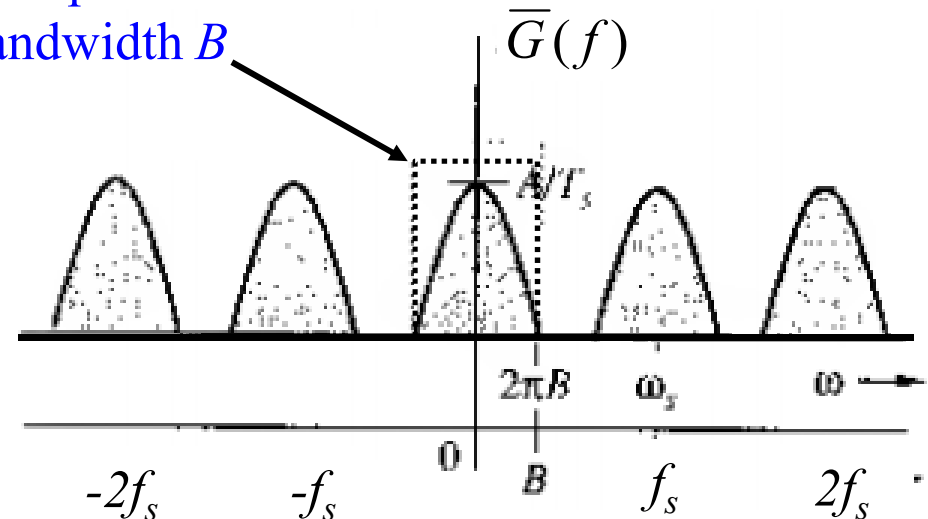
Sampling Theorem

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- For a general signal $g(t)$

$$f_s > 2B$$



Low pass filter of
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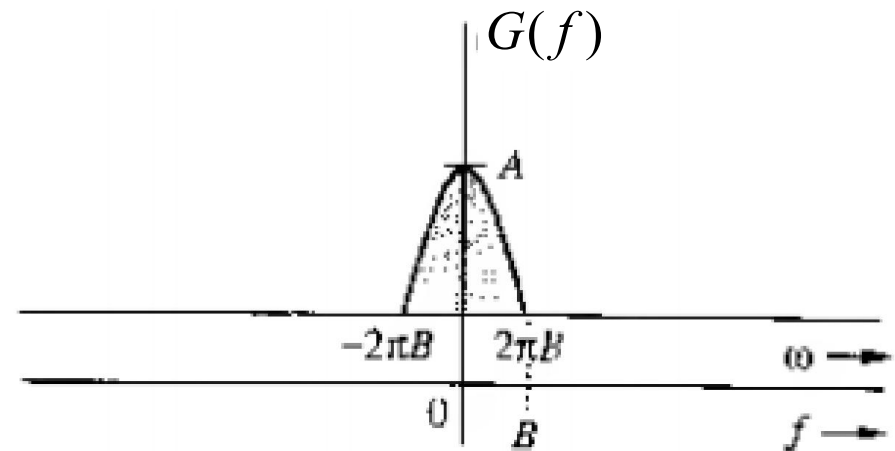
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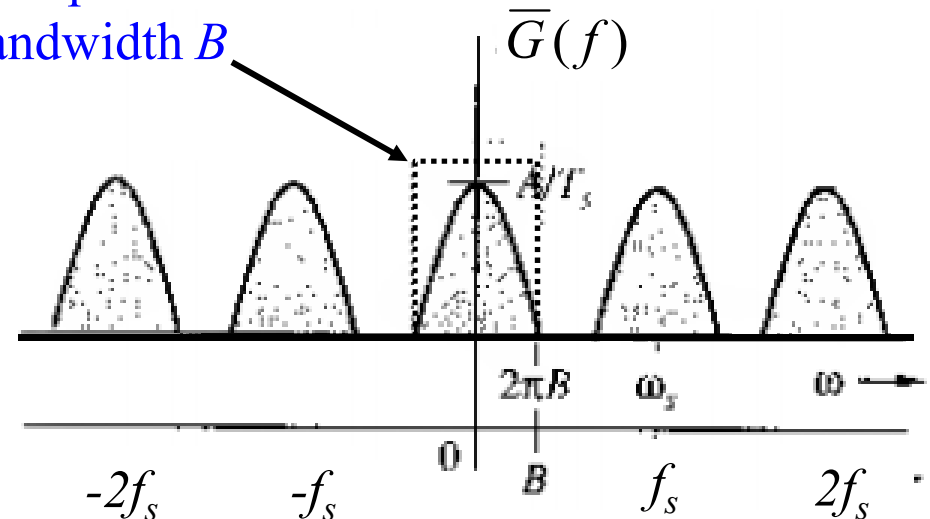
$$f_s > 2B$$

- If there is no impulse at $\pm B$, we can use $f_s = 2B$. Therefore,

$$f_s \geq 2B$$



Low pass filter of bandwidth B



Sampling Theorem

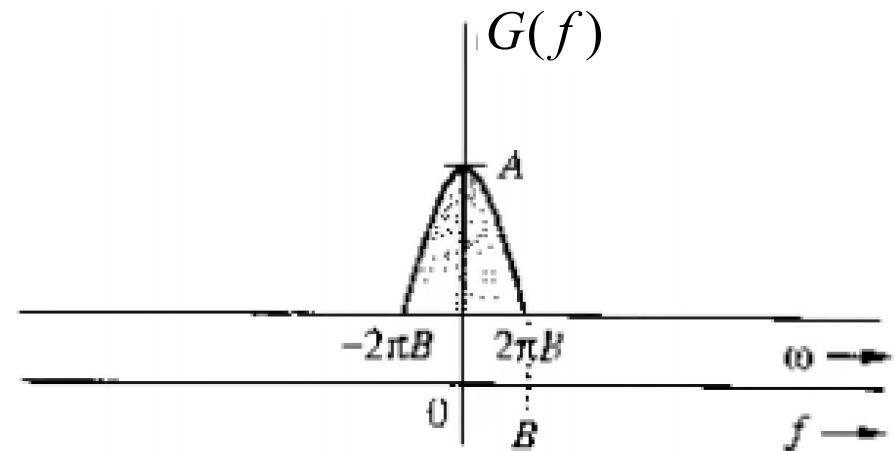
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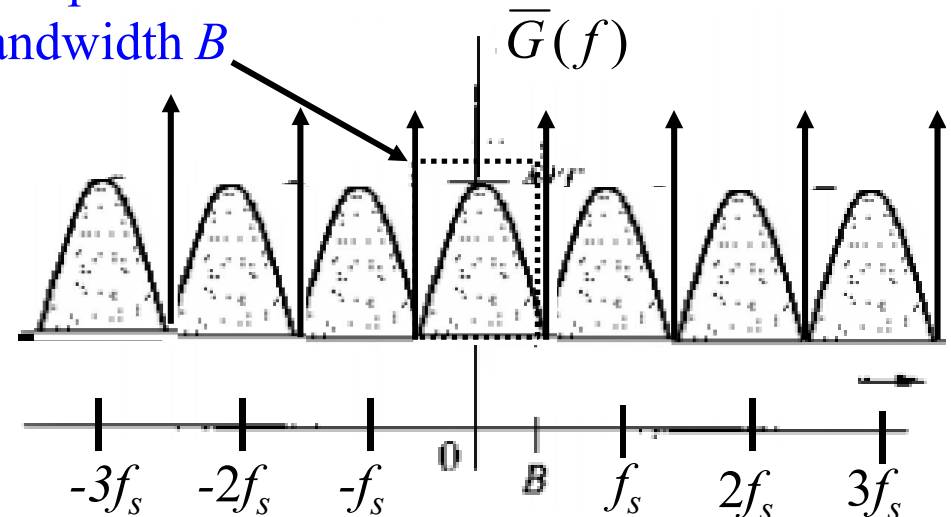
- If there is no impulse at $\pm B$, we can use $f_s = 2B$. Therefore,

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- If there is IMPULSE at $\pm B$, we can NOT use $f_s = 2B$, because overlap will occur. Therefore, $f_s > 2B$

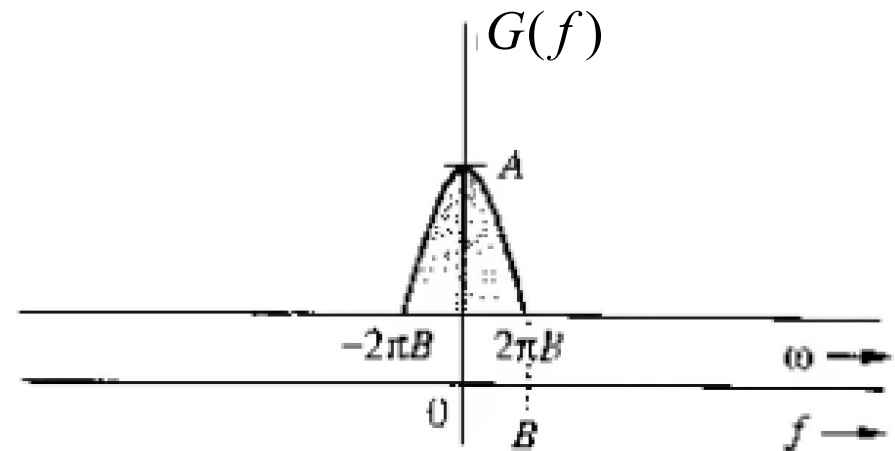


Low pass filter of bandwidth B

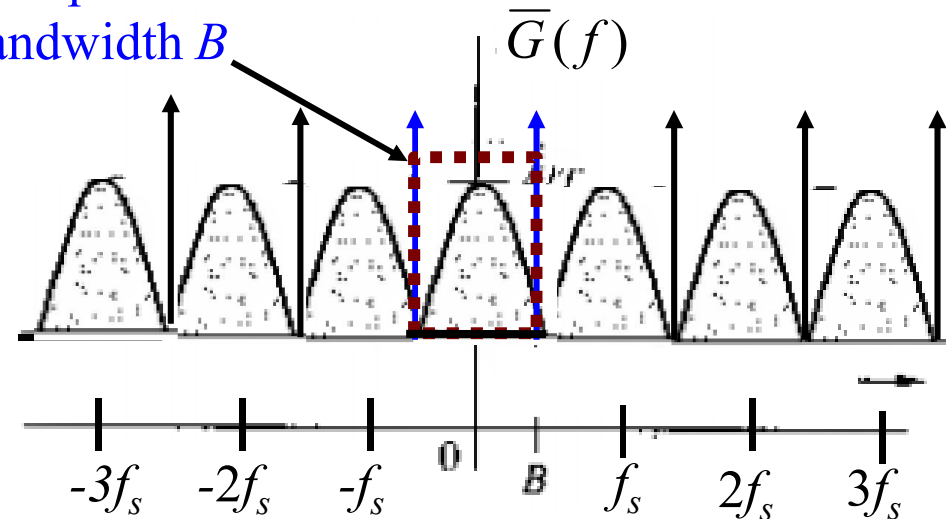


Sampling Theorem

- In this case we cannot isolate one cycle from $\overline{G}(f)$ if $f_s = 2B$

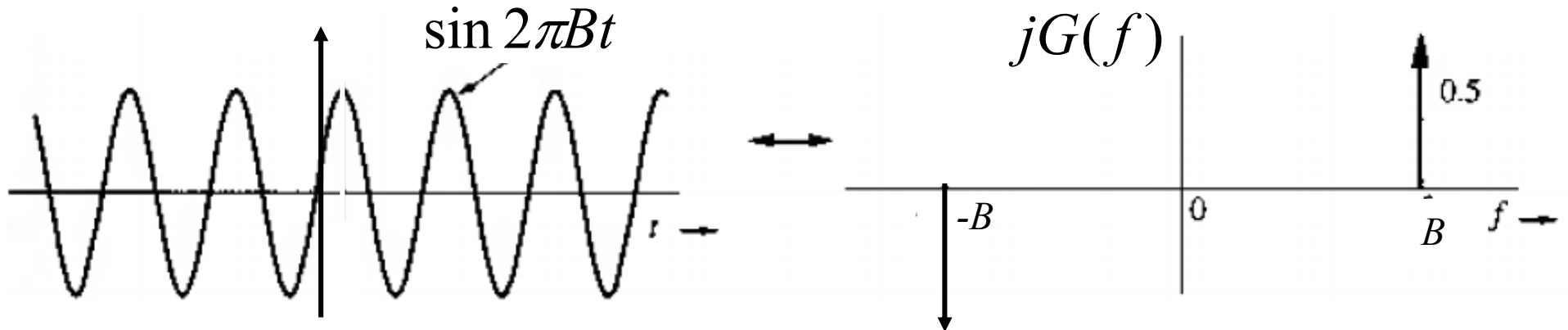


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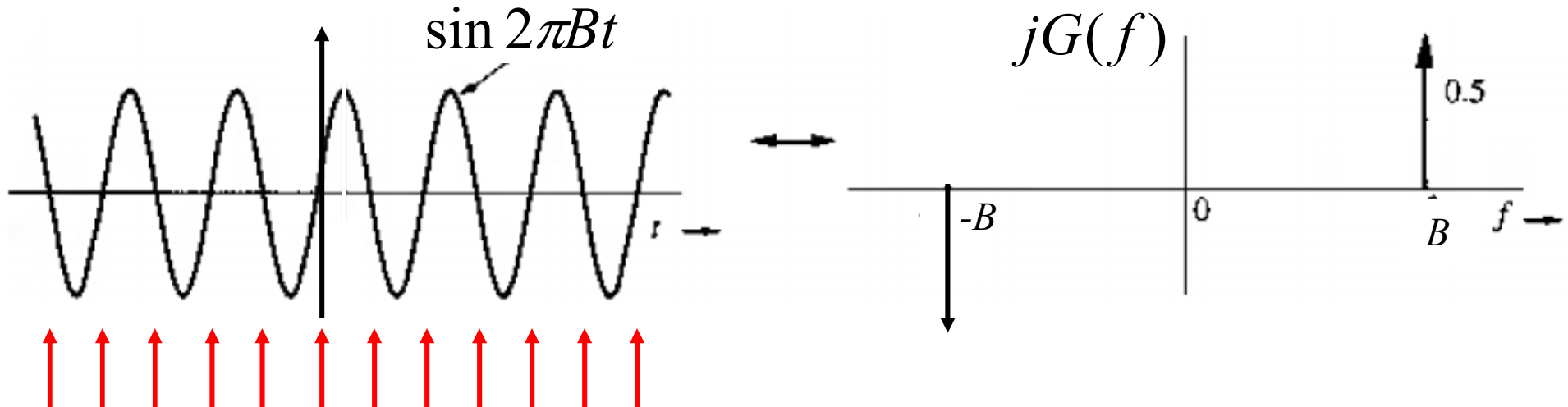
Sampling Theorem

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- Example of sine web of frequency B



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- Example of sine web of frequency B



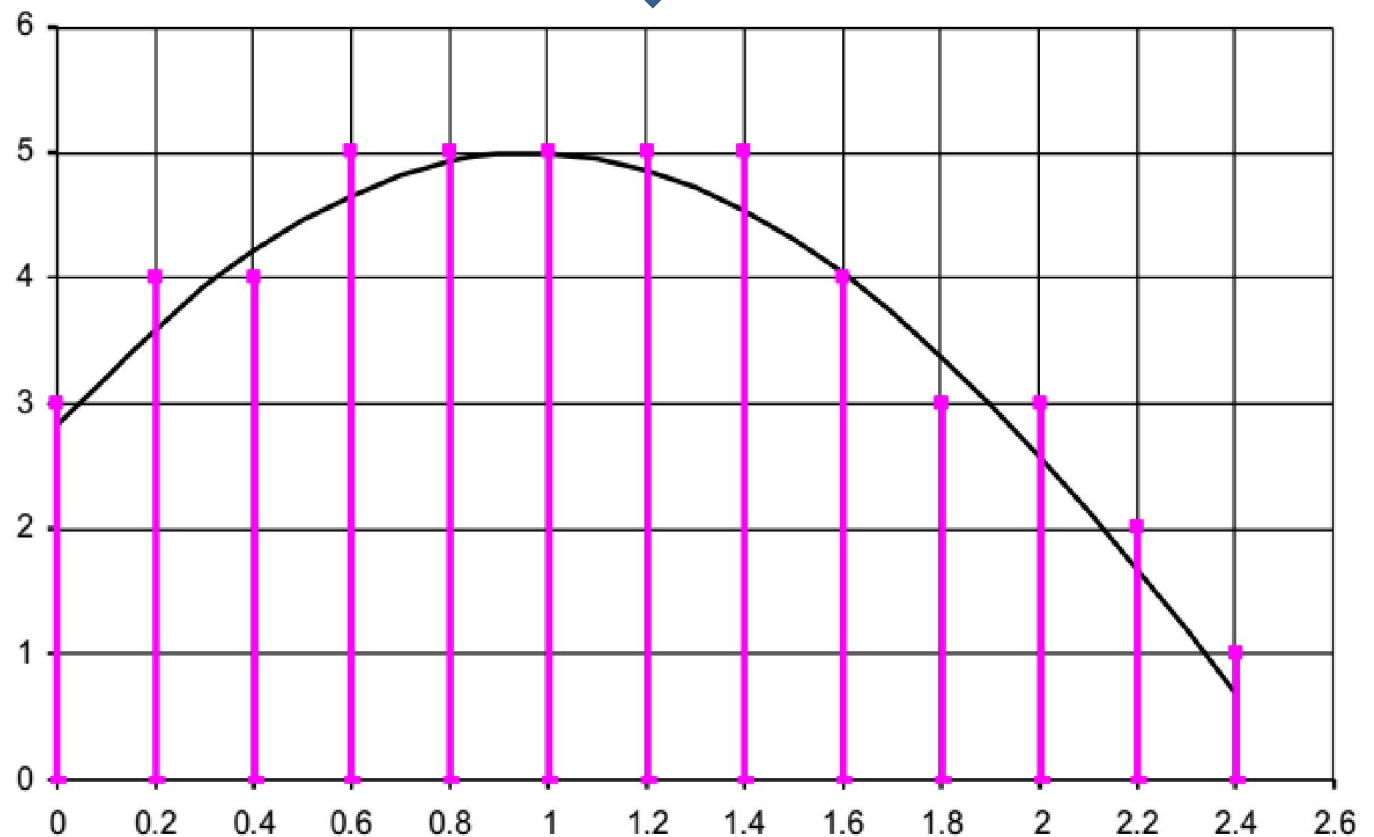
Sampling at the rate of $2B$ at these points
will give constant values (zeros)

Signal Reconstruction from Uniform Samples

- Reconstruction objective

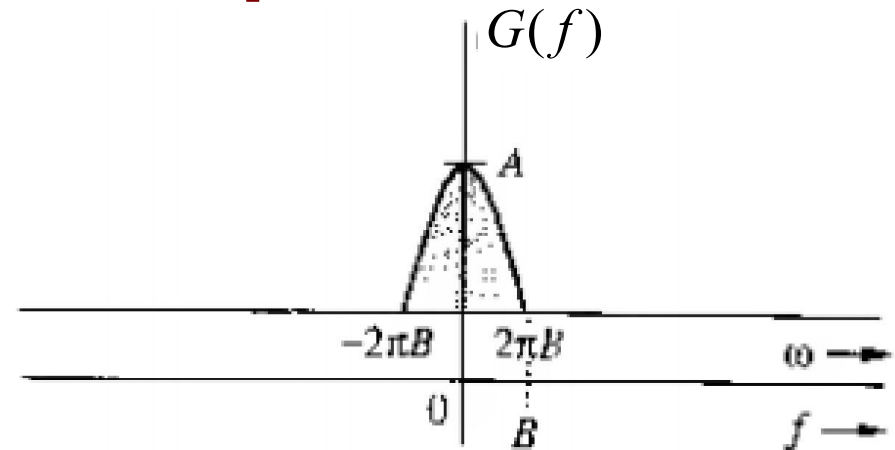
Need to
interpolate
intermediate
values

2.823212	3.58678	4.207355	4.660195	4.927249	
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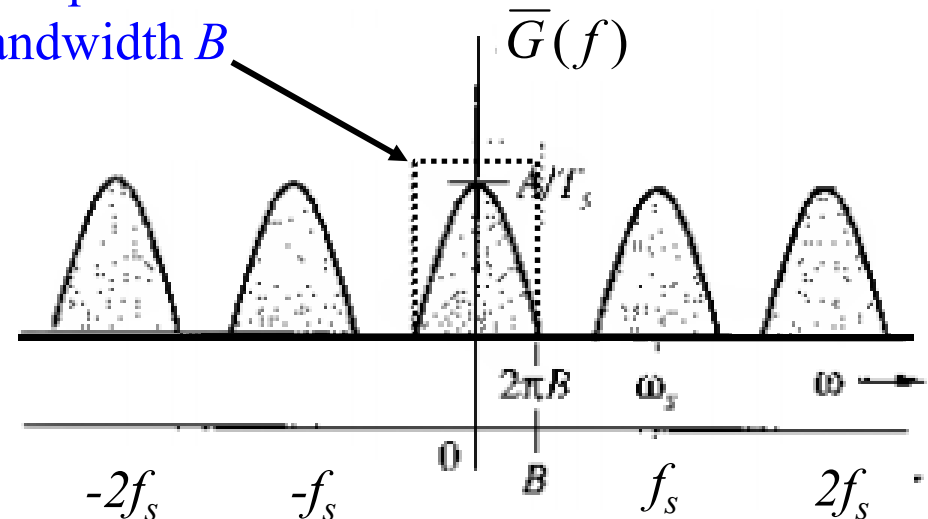


Signal Reconstruction from Uniform Samples

- We proved theoretically exact reconstruction is possible provide
 - the signal is sampled at rate $f_s > 2B$
 - The sampled signal is passed through an ILPF with bandwidth B



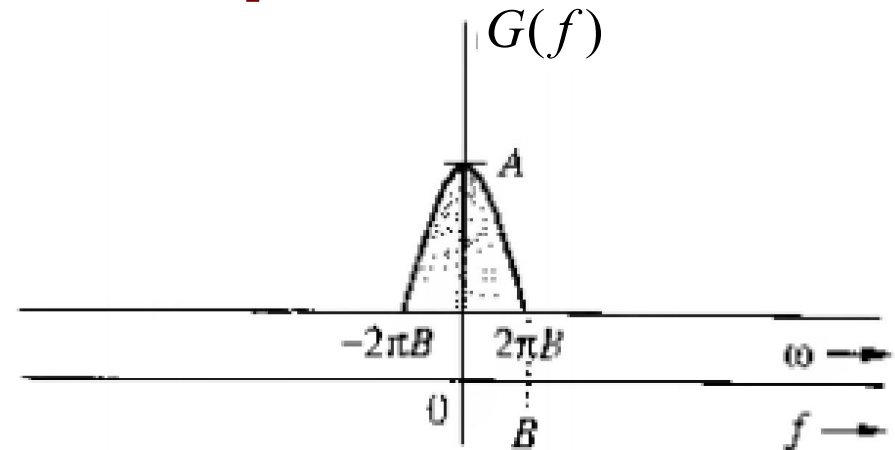
Low pass filter of bandwidth B



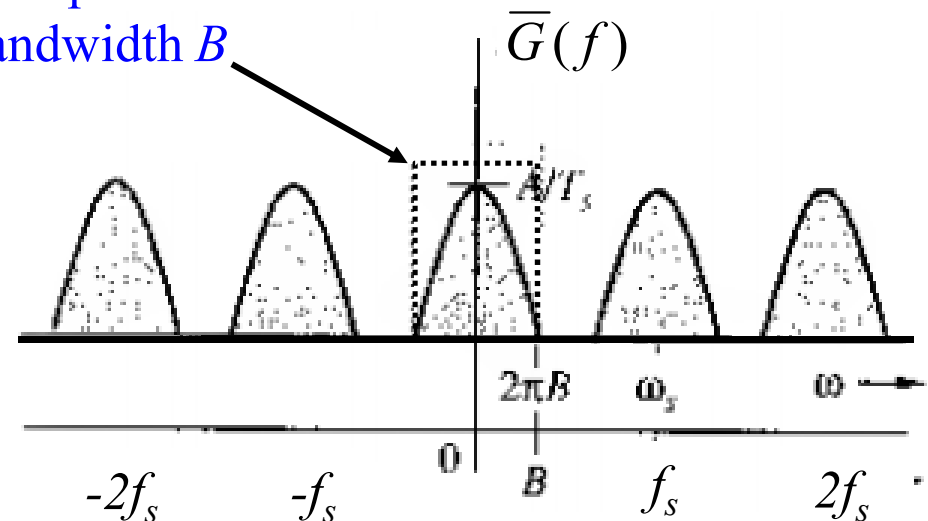
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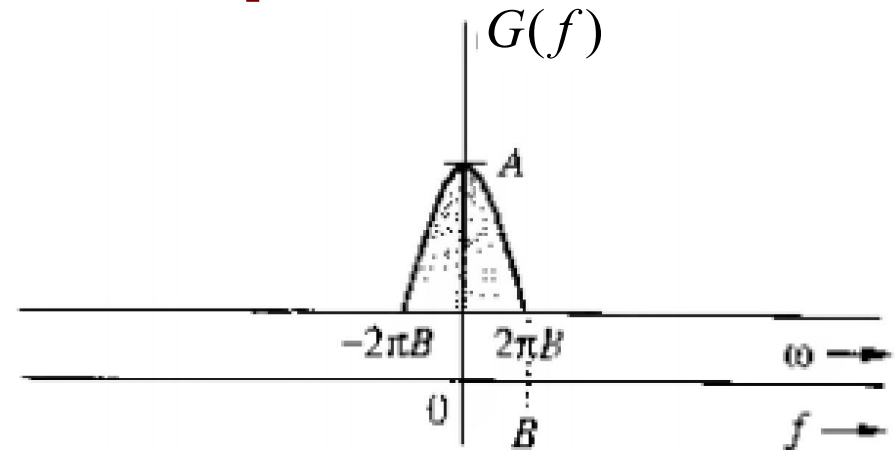


Low pass filter of bandwidth B



Signal Reconstruction from Uniform Samples

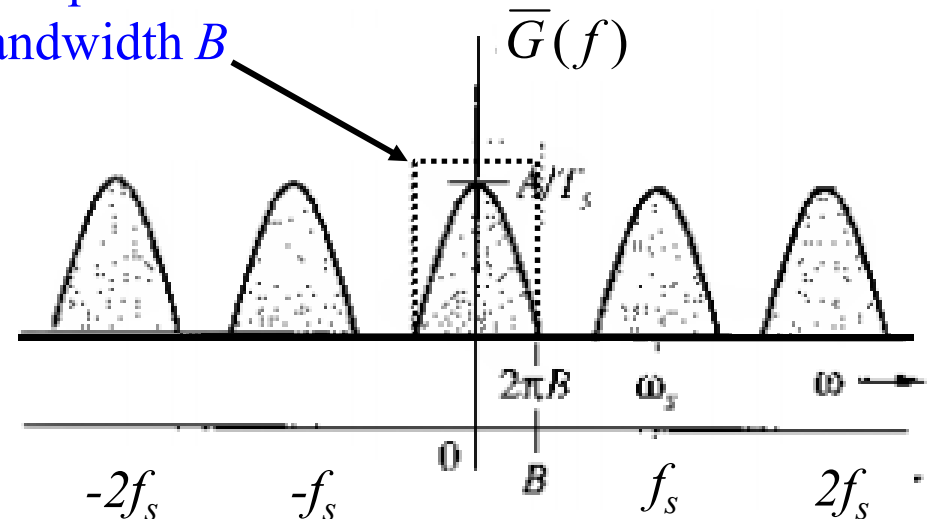
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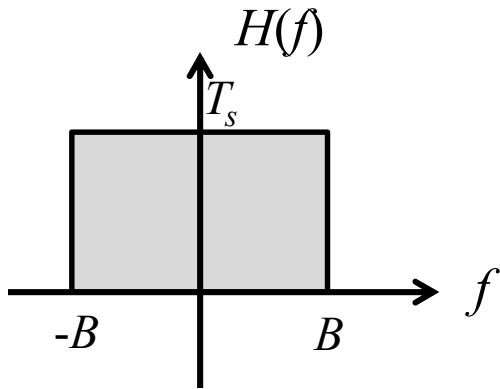
$$H(f) = T_s \Pi\left(\frac{f}{2B}\right)$$

Low pass filter of bandwidth B



Signal Reconstruction from Uniform Samples

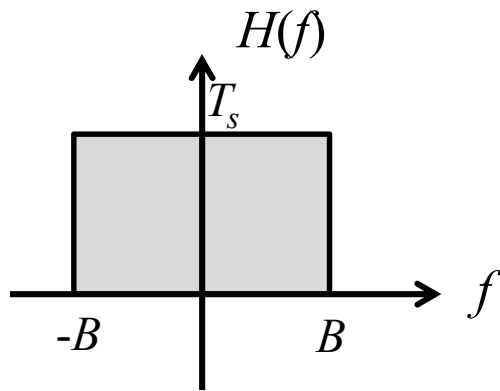
- To recover $g(t)$, the transfer function will be $H(f) = T_s \Pi(\frac{f}{2B})$



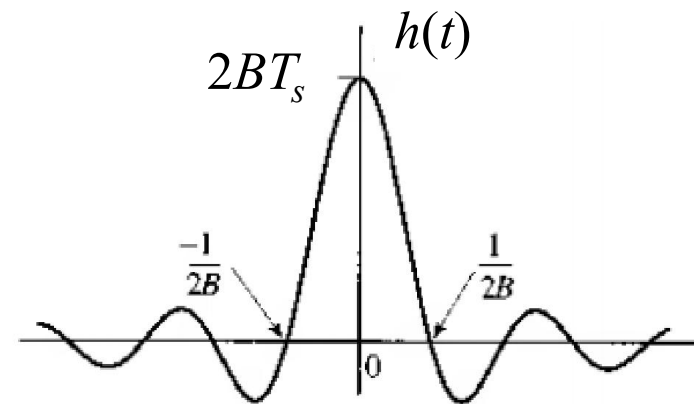
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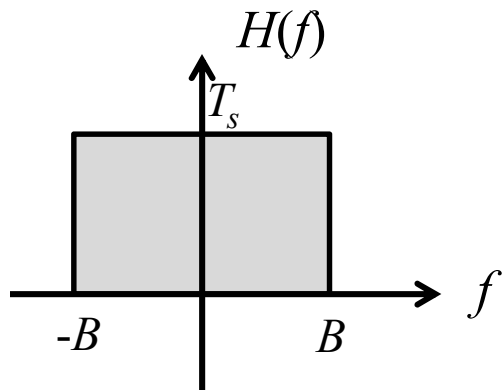


$$h(t) = 2BT_s \text{sinc}(2\pi Bt)$$

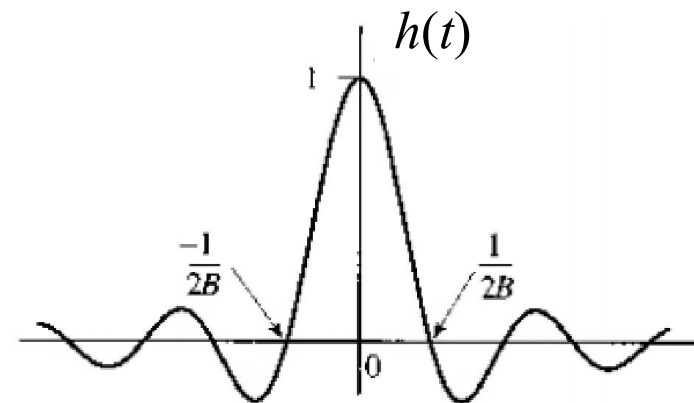
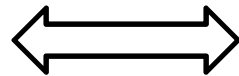
Impulse response

Signal Reconstruction from Uniform Samples

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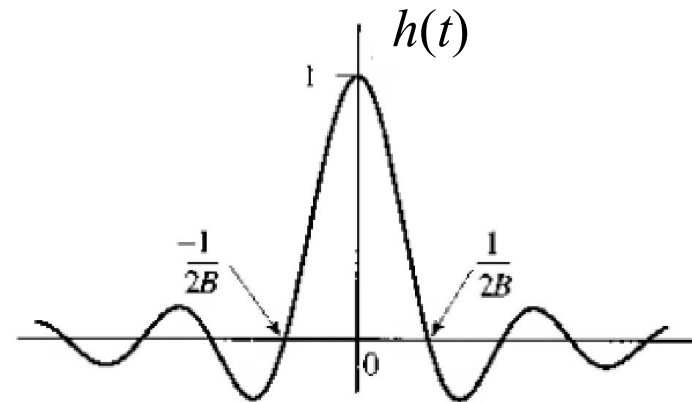
$$\begin{aligned} h(t) &= 2BT_s \text{sinc}(2\pi Bt) \\ &= \text{sinc}(2\pi Bt) \end{aligned}$$

Assuming Nyquist sampling interval $T_s = \frac{1}{2B}$

Signal Reconstruction from Uniform Samples

- To recover $g(t)$, the transfer function will be $H(f) = T_s \Pi(\frac{f}{2B})$

Note: $h(t) = 0$ at Nyquist sampling instant $\pm \frac{n}{2B}$



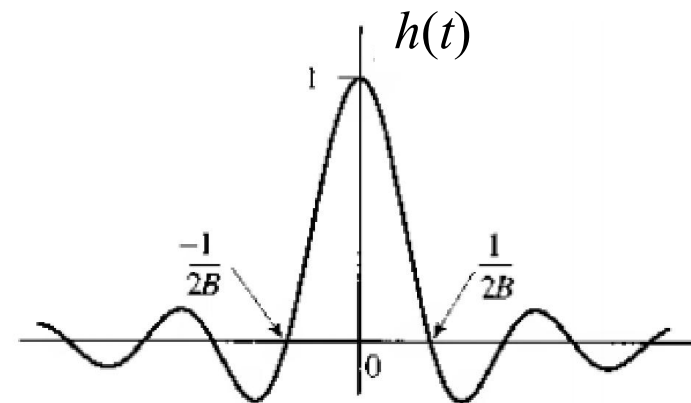
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Impulse response

Signal Reconstruction from Uniform Samples

- To recover $g(t)$, the transfer function will be $H(f) = T_s \Pi(\frac{f}{2B})$

If we pass $\bar{g}(t)$ through an LTI with this impulse, the output will be $g(t)$.



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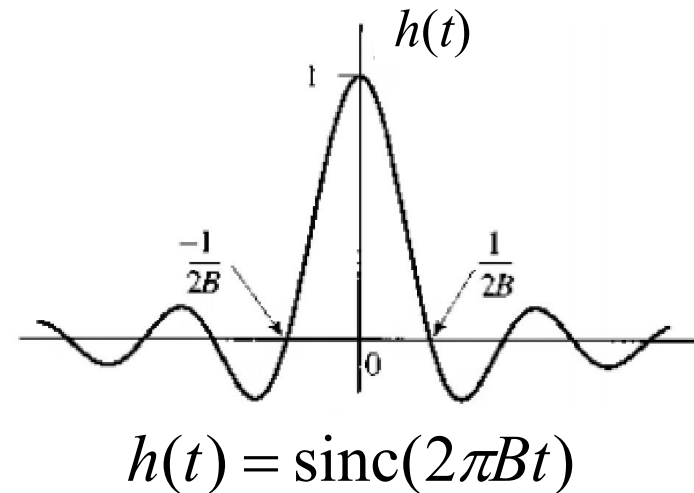
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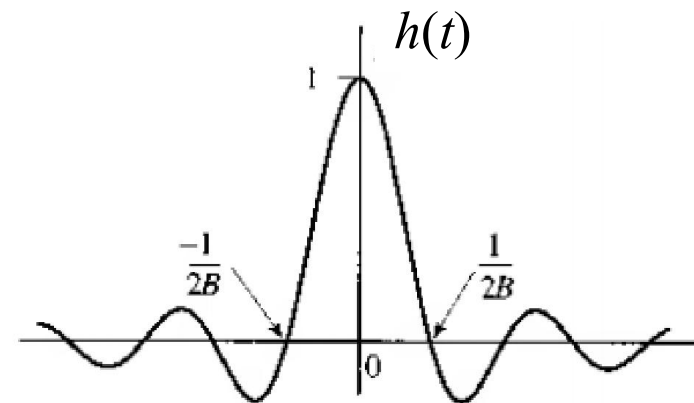


Signal Reconstruction from Uniform Samples

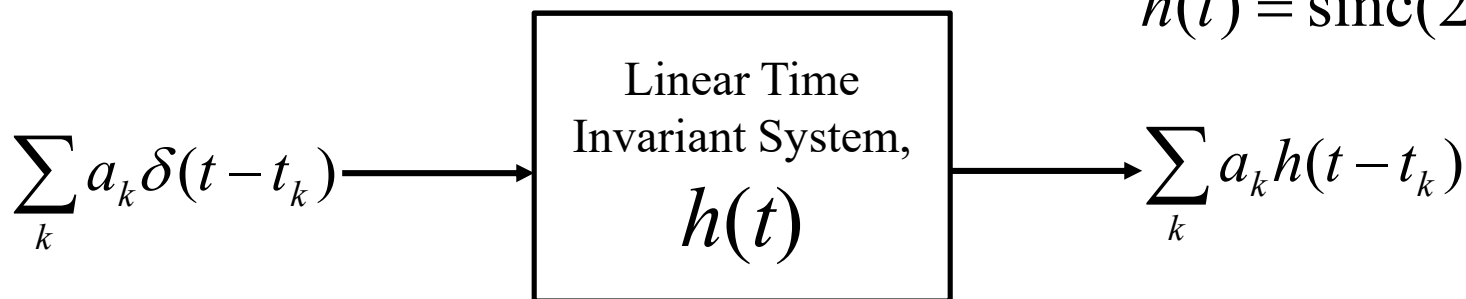
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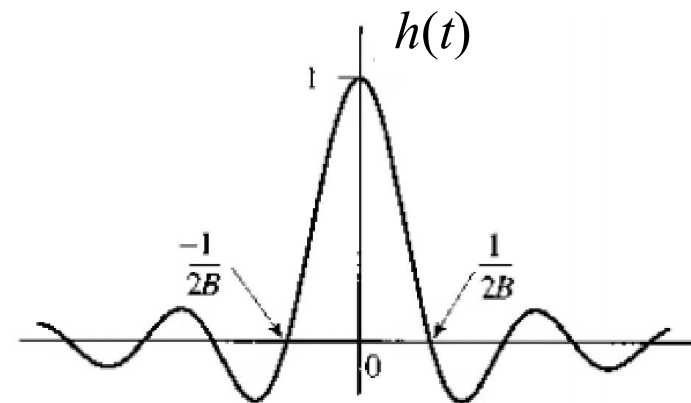


Signal Reconstruction from Uniform Samples

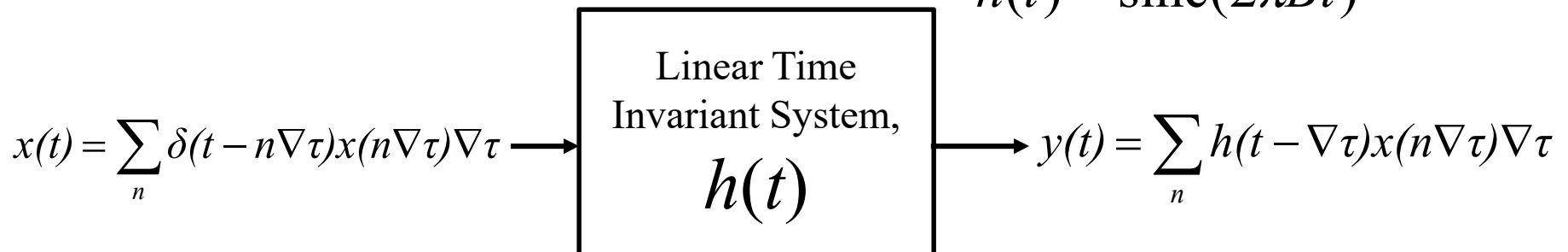
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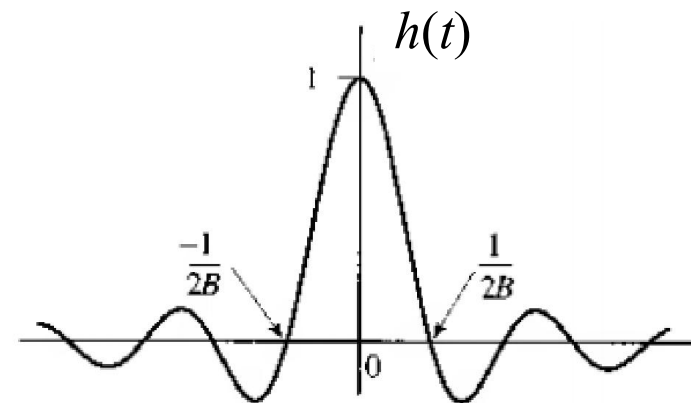


Signal Reconstruction from Uniform Samples

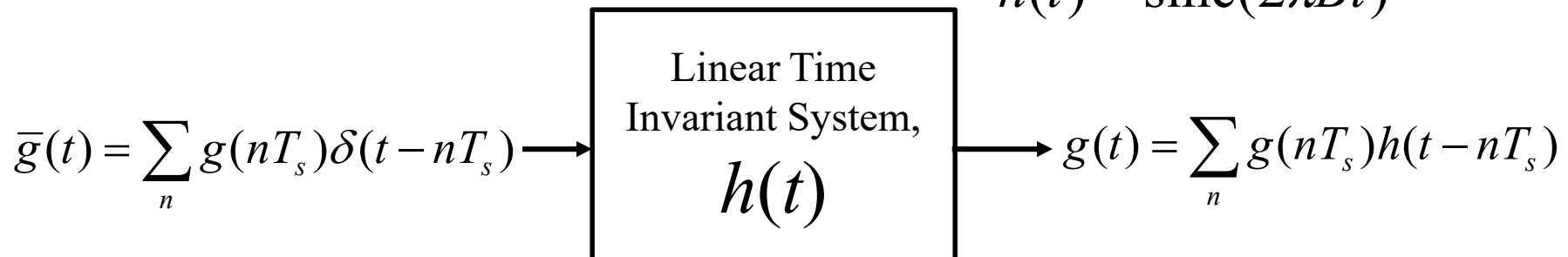
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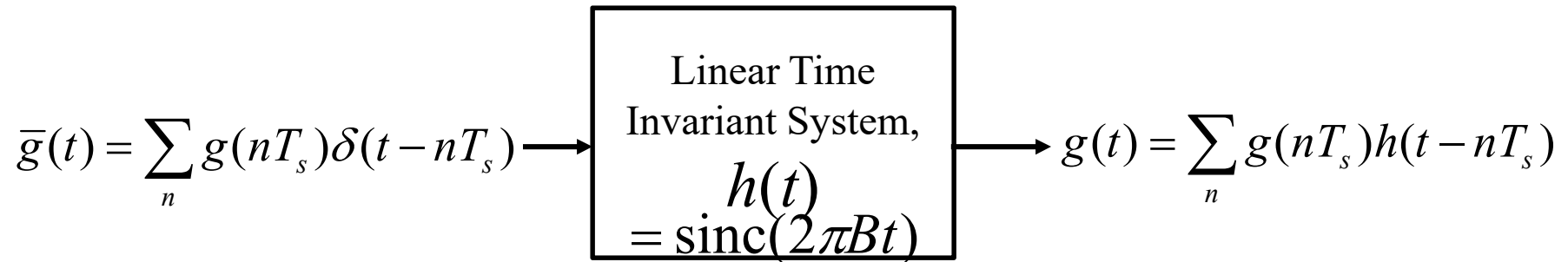
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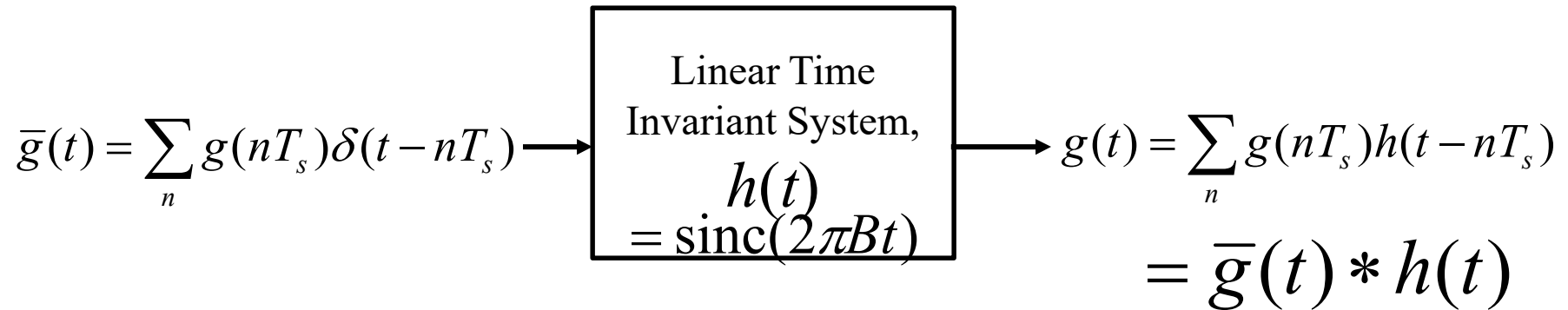
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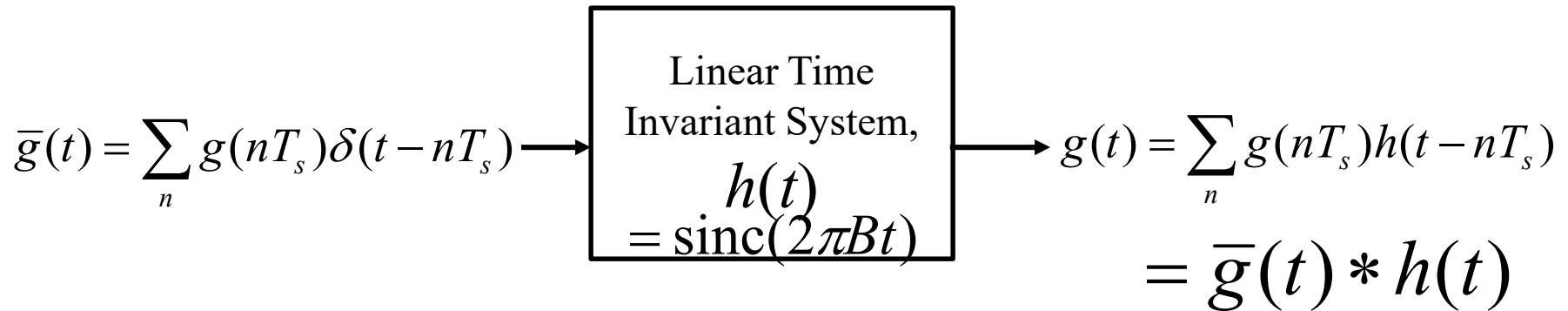
Signal Reconstruction from Uniform Samples



Signal Reconstruction from Uniform Samples

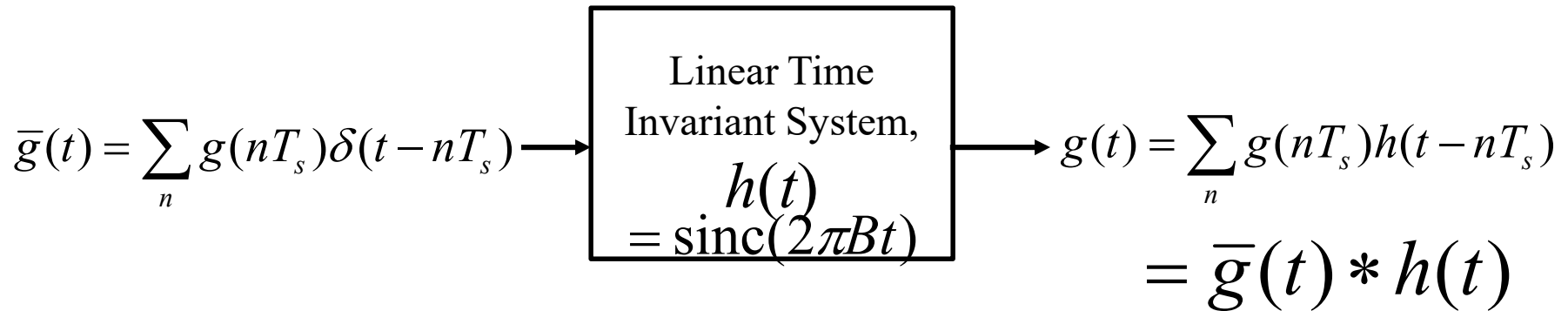


Signal Reconstruction from Uniform Samples



$$\begin{aligned} g(t) &= \sum_n g(nT_s) h(t - nT_s) \\ &= \sum_n g(nT_s) \text{sinc}[2\pi B(t - nT_s)] \\ &= \sum_n g(nT_s) \text{sinc}(2\pi Bt - n\pi) \end{aligned}$$

Signal Reconstruction from Uniform Samples



$$g(t) = \sum_n g(nT_s) \text{sinc}(2\pi Bt - n\pi)$$

Interpolation formula to find the value $g(t)$ at arbitrary time t .

Signal Reconstruction from Uniform Samples

Significance of Interpolation formula

$$g(t) = \sum_n g(nT_s) \text{sinc}(2\pi Bt - n\pi)$$

